

Benchmarking Quantum Simulation Methods

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Quantum Phase Estimation (QPE) is a cornerstone algorithm for fault-tolerant quantum computation, especially for electronic structure calculations of chemical systems. To accommodate the diverse characteristics of quantum chemical systems, numerous variants of QPE have been developed, each with distinct qubit and gate cost implications. In this paper, we quantify the impact of three key parameters on the overall quantum resource costs for the QPE algorithm: the choice between trotterization and qubitization, the use of molecular orbitals versus plane-wave basis-sets, and the selection of the fermion-to-qubit encoding scheme. From this, we establish clear performance trade-offs and delineate specific parameter regimes that minimize resource costs for relevant molecular systems. When performing phase estimation on large molecules in the fault-tolerant setting, we found the first-quantized qubitization circuit using the plane-wave basis to be the most efficient, with a gate cost scaling of $\tilde{O}([N^{4/3}M^{2/3} + N^{8/3}M^{1/3}]/\epsilon)$ for a system of N electrons and M orbitals, which is the best known scaling to date. On the other hand, when only noisy intermediate-scale or near-term fault-tolerant systems are available, the phase estimation of small molecules can be performed with gate cost of $\mathcal{O}(M^7/\epsilon^2)$ via trotterization in the MO basis. Furthermore, we provide numerical estimations of qubit and T gate costs required to perform QPE for several real-world molecular systems under these different parameter choices.

Introduction

Significant advancements in quantum computing have opened up unprecedented opportunities for solving complex problems, particularly in the realm of quantum chemistry. The ability to accurately model and simulate molecular systems on quantum computers holds the potential to revolutionize drug discovery, materials science, and fundamental chemical research. Central to these simulations is the effective implementation of quantum chemical Hamiltonians onto quantum circuits, a task primarily addressed by two prominent methodologies: trotterization [1, 2] and the qubitization approach [3, 4]. As the major costs of most quantum simulation algorithms involves repeated applications of these methods [1–10], it is crucial to have efficient implementations of both.

A key advantage of trotterization lies in its conceptual simplicity and its minimal use of ancilla qubits. However, its computational cost typically scales polynomially with the inverse of the desired error. Furthermore, tight theoretical error bounds can be challenging to obtain, often being several times larger than experimental error calculations [10]. Consequently, trotterization is often considered more suitable for small to medium-sized system on current noisy intermediate-scale quantum (NISQ) devices or near-term partially fault-tolerant architectures. On the other hand, qubitization methods fare better in terms of their error scaling. As an example, Hamiltonian evolution based on qubitization [5] achieves computational costs with logarithmic dependence on the reciprocal of the error parameter. This comes at a cost of

a more complex circuit constructions, often with multiple Quantum Read-Only Memory (QROM) implementations, and requiring additional ancilla qubits. This makes qubitization-based approaches particularly well-suited for larger systems in future fault-tolerant quantum computers.

To accurately quantify the costs associated with both methods, it is essential to consider the various parameters affecting the computational costs and qubit requirements. The major parameters include the fermion-to-qubit encoding and the basis set used for the calculations. For fermion-to-qubit encodings, the Jordan-Wigner [11] encoding has been the standard for previous cost analyses for trotterization [2, 12, 13] and qubitization [6–8]. An obvious step to improve qubit costs is to use an encoding that leverages the particle conservation symmetry inherent in quantum chemical Hamiltonians. To this end, the first-quantized encoding [14–16] offers an improvement in qubit cost in the $N \ll M$ regime, scaling as $\mathcal{O}(N \log M)$ compared to $\mathcal{O}(M)$ for the Jordan-Wigner encoding, where M is the number of orbitals and N is the number of electrons. For second-quantized encodings, the sorted-list encoding [17] shows reasonable theoretical improvements over the Jordan-Wigner encoding, achieving $\mathcal{O}(N \log M)$ scaling in both qubit count and gate cost to implement a fermionic operator.

Another critical parameter affecting the costs associated with trotterization and qubitization is the choice of the basis set. The molecular orbital (MO) basis provides a compact representation of the chemical system, requiring a relatively smaller number of orbitals compared to the plane-wave basis. Since MOs are constructed as eigenvectors of mean-field Hartree-Fock calculations, their associated orbital energies simplify the implemen-

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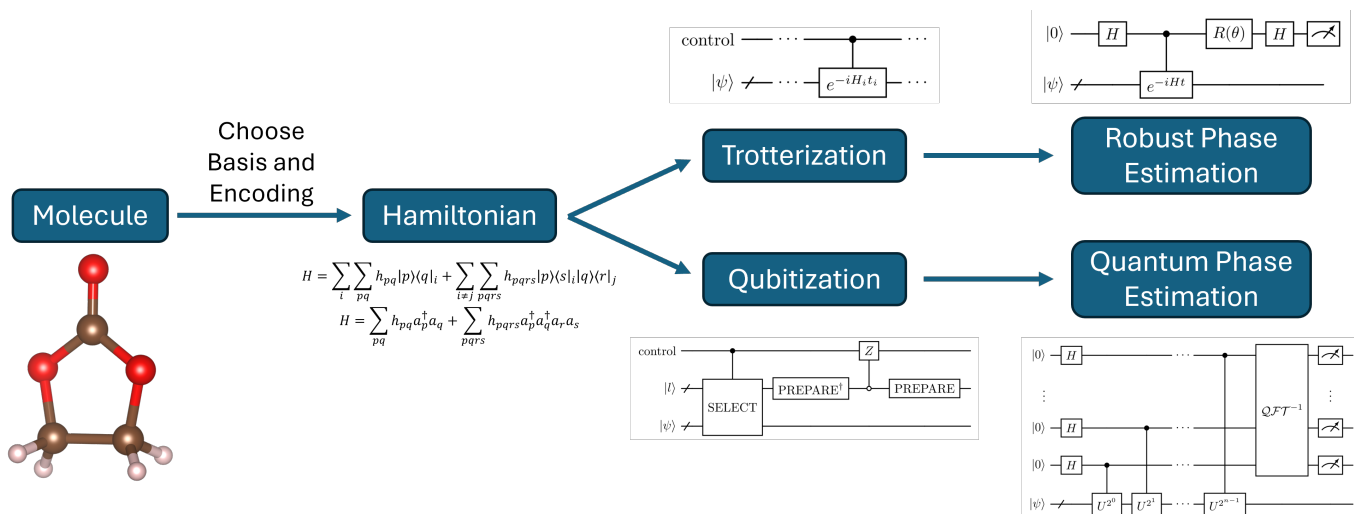


Figure 1: Workflow for an energy estimation algorithm using quantum computers, using either trotterization or qubitization to effectively implement the quantum chemical Hamiltonian into the quantum circuit.

tation of active space methods for ground-state calculations. This technique removes certain orbitals and electrons from the quantum calculation, with a small trade-off in ground-state accuracy, by assuming that orbitals associated with the lowest and highest energies are always occupied or unoccupied, respectively, rather than being in a superposition. However, the MO basis comes with its own trade-offs. For qubitization, the arbitrary nature of the h_{pq} and h_{pqrs} coefficients necessitates extensive use of QROMs for their implementation. While these costs can be somewhat reduced through rank-reduction techniques [6–8], this involves expensive classical preprocessing.

On the other hand, plane-wave basis sets possess analytical coefficients that can be efficiently computed within quantum circuits. This significantly benefits first-quantized qubitization circuits, as they rely on QROMs to a lesser extent compared to MO basis sets. In fact, this first-quantized qubitization circuit has the most efficient computational complexity scaling with respect to the number of plane waves to date [15, 16]. Moreover, the kinetic term of the Hamiltonian is diagonal in the plane-wave basis, and the potential terms are diagonal in the plane-wave dual basis. This greatly benefits first-quantized trotterization circuits, as all Hamiltonian terms can be grouped into two unitaries: one for kinetic terms and another for potential terms, with efficient quantum Fourier transform circuits enabling conversion between the plane-wave and its dual basis. This profound advantage for first-quantized approaches contrasts with second-quantized trotterization and qubitization circuits, where the analytical structure of plane-wave coefficients offers only minor benefits. This advantage also comes at the trade-off of requiring a much larger number of plane waves, typically around 1000 times the number of MO orbitals, to achieve the same level of precision [16]. Active space techniques cannot be easily used to reduce the

number of electrons in this context; instead, pseudopotentials are often employed to replace core electrons with approximate potentials.

In this paper, we present a comprehensive cost analysis of trotterization and qubitization techniques for ground-state energy estimation. Our analysis elucidates the impact of different encoding schemes and basis sets on both computational cost and qubit count, providing a quantitative characterization of their respective trade-offs and identifying regimes where each of encoding scheme provides a computational advantage. Specifically, we integrate trotterization with robust phase estimation (RPE) [2] and the qubitization with quantum phase estimation (QPE) [18] in order to estimate the resource requirements of such algorithms in several model chemical systems.

Results

Our comprehensive cost analysis of trotterization and qubitization techniques for quantum simulation systematically investigates the impact of different encoding schemes and basis sets on both gate cost and qubit count. Here we summarize our results.

Due to the significant differences in algorithm implementations between the MO and plane-wave bases, we present separate discussions of their cost scalings for both the trotterization and qubitization implementations. Notably, the plane-wave basis trades its larger basis-set size for a significantly improved gate cost scaling. For the RPE algorithm with trotterization, the gate and qubit cost scalings are provided in Theorem 1 for the MO basis and Theorem 2 for the plane-wave basis. On the other hand, the cost analysis for the QPE algorithm using qubitization is shown in Table IV for the MO basis and Theorem 3 for the plane-wave basis.

For the trotterization-based algorithm in the MO basis, we found that the sorted-list encoding obtains a qubit

cost advantage in the $N \ll M$ regime. However, the advantage in gate cost is more nuanced: while it offers better Clifford gate scaling, its T gate scaling is worse. Despite this, we still expect an advantage in total quantum resource costs for the sorted-list encoding in the $N/M \rightarrow 0$ limit. In contrast, for the plane-wave basis, the sorted-list encoding maintains parity with the first-quantized encoding in terms of the gate and qubit cost complexities and provides an advantage over the Jordan-Wigner encoding.

For the qubitization-based algorithm in the MO basis, we found similar gate and qubit costs between the sorted-list and the Jordan-Wigner encodings, as the dominant terms are independent of the encoding choice. Conversely, the first-quantized encoding still holds an advantage over the sorted-list encoding in the plane-wave basis due to the inherently lower 1-norm scaling of the first-quantized Hamiltonian.

The two bases discussed also have different applicability for different systems. The MO basis has been the conventional choice for molecular systems due to its localized nature. On the other hand, the periodicity of the plane-wave basis is more suitable for periodic systems. Despite this, the tighter bounds provided by the plane-wave basis may still be preferable over the MO basis for molecular systems. In such cases, the molecule must be placed in a simulation large enough to minimize interaction with its periodic images.

Throughout the paper, we adopt the notation $\tilde{\mathcal{O}}(f) = \mathcal{O}(f \text{ polylog}(M, N, \lambda, \varepsilon))$, where λ is the Hamiltonian 1-norm, and ε is the precision parameter.

Trotterization

Our implementation and cost analysis of the trotterization circuits uses the single-ancilla phase estimation algorithm called robust phase estimation (RPE) [2]. The single ancilla requirement of the RPE algorithm pairs well with the low ancilla overhead of trotterization. This algorithm requires the implementation of controlled e^{-iHt} gates, which we approximate using various Trotter formulas. The Hamiltonian is first decomposed into a linear combination of Hermitian operators $H = \sum_{l=0}^{L-1} h_l H_l$, where each Hermitian H_l is chosen such that $e^{-ih_l H_l t_l}$ can be efficiently implemented as a quantum circuit for arbitrary time steps t_l .

The efficiency of implementing these exponential operators, and thus the overall Trotter simulation, is highly dependent on the chosen fermion-to-qubit encoding. The sorted-list encoding represents the latest attempt in obtaining the optimal $\mathcal{O}(N \log M)$ qubit scaling for simulating a second-quantized particle-conserving Hamiltonians. With its distinct qubit advantage over the Jordan-Wigner in the $N \ll M$ limit, the objective here is to also provide similar or improved gate complexities. Historically, this has proven challenging, as earlier encodings that offered qubit savings [19–23] either incurred severe trade-offs in gate complexities or did not achieve the optimal qubit scaling.

Trotterization for the MO basis. We begin our analysis with the MO basis, where the h_{pq} and h_{pqrs} coefficients of the Hamiltonians are arbitrary, subject only to the Hermiticity of the Hamiltonian. Furthermore, we assume the Hamiltonian is decomposed into L Hermitian terms, and its 1-norm is $\lambda = \sum_{l=0}^{L-1} |h_l|$. Theorem 1 formalizes the gate and qubit cost scalings to perform the RPE algorithm that obtains the ground-state energy up to precision ε .

Theorem 1. *The ground-state energy of a Hamiltonian H can be calculated to precision ε using the RPE algorithm. For the sorted-list encoding, the gate costs (both Clifford and T gates) scale as:*

- $\mathcal{O}(N \log M [5^{p/2-1} L^{2+1/p} / \varepsilon^{1+1/p}])$ when using the deterministic p^{th} order trotter formula [24],
- $\mathcal{O}(N \log M [\lambda^2 / \varepsilon^2])$ when using the random qDRIFT formula [25], with $\lambda = \sum_{l=0}^{L-1} |h_l|$,
- $\mathcal{O}(N \log M [5^{p/2-1} L_{\text{det}}^{2+1/p} / \varepsilon^{1+1/p} + \lambda_{\text{rand}}^2 / \varepsilon^2])$ when using a partially random approach [2], where the Hamiltonian is partitioned into deterministic (H_{det}) and random (H_{rand}) parts:

$$e^{-iHt} = e^{-iH_{\text{det}}t} e^{-iH_{\text{rand}}t}.$$

For the partially random approach, L_{det} and λ_{rand} only refer to the deterministic and random parts, respectively. The qubit costs scales identically across these three strategies at $\mathcal{O}(N \log M)$.

Table I compares the gate and qubit costs of the results in Theorem 1 with those of the Jordan-Wigner encoding. The sorted-list encoding has its qubit requirement scale as $\mathcal{O}(N \log M)$ compared to $\mathcal{O}(M)$ for Jordan-Wigner, providing a significant advantage in the low electron-filling regime ($N \ll M$) prevalent in quantum chemical Hamiltonians. This qubit saving does come at a cost of increased T gate complexity. For instance, the time evolution of each Hermitian fermionic operator costs $\mathcal{O}(N \log M)$ Clifford and T gates in the sorted-list encoding. In contrast, while the Jordan-Wigner encoding has worse Clifford gate scaling at $\mathcal{O}(M)$, it only requires $\mathcal{O}(1)$ T gates for every Pauli string.

Given that the T gates are significantly more expensive to implement than Clifford gates, the Jordan-Wigner encoding will generally offer an advantage for typical Hamiltonian simulations where the precision is within chemical accuracy. However, in the $N/M \rightarrow 0$ limit, the sorted-list encoding is expected to regain its advantage due to the logarithmic dependence on M for both space and time resources. The balanced logarithmic dependence on M is a rare property in fermion-to-qubit encodings, as shown in Table II, with the polylog encoding [22] being the only other encoding exhibiting similar dependence on M . Such low N/M ratios in the MO basis may be encountered in scenarios demanding exceptionally high accuracy, such as when studying Van der Waals interactions or hyperfine structures.

Table I: Gate and qubit costs comparison between the sorted-list and Jordan-Wigner encodings for obtaining the ground-state energy of H with precision ε using the RPE algorithm for the three different Trotter product formulas. For the sorted-list encoding, the Clifford gate and the T gate costs have identical scalings. For the partially random approach, L_{det} and λ_{rand} only refer to the deterministic and random parts, respectively. We typically have $L = \mathcal{O}(M^4)$ and $\lambda = \mathcal{O}(M) \sim \mathcal{O}(M^3)$.

Formula	Sorted-List	Jordan-Wigner	
	Clifford/T gate	Clifford gate	T gate
Deterministic p^{th} order	$\mathcal{O}(N \log M [5^{p/2-1} L^{2+1/p} / \varepsilon^{1+1/p}])$	$\mathcal{O}(M [5^{p/2-1} L^{2+1/p} / \varepsilon^{1+1/p}])$	$\mathcal{O}(5^{p/2-1} L^{2+1/p} / \varepsilon^{1+1/p})$
Random qDRIFT	$\mathcal{O}(N \log M [\lambda^2 / \varepsilon^2])$	$\mathcal{O}(M [\lambda^2 / \varepsilon^2])$	$\mathcal{O}(\lambda^2 / \varepsilon^2)$
Partially random	$\mathcal{O}(N \log M [5^{p/2-1} L_{\text{det}}^{2+1/p} / \varepsilon^{1+1/p} + \lambda_{\text{rand}}^2 / \varepsilon^2])$	$\mathcal{O}(M [5^{p/2-1} L_{\text{det}}^{2+1/p} / \varepsilon^{1+1/p} + \lambda_{\text{rand}}^2 / \varepsilon^2])$	$\mathcal{O}(5^{p/2-1} L_{\text{det}}^{2+1/p} / \varepsilon^{1+1/p} + \lambda_{\text{rand}}^2 / \varepsilon^2)$
Qubit cost	$\mathcal{O}(N \log M)$		$\mathcal{O}(M)$

Table II: Clifford, T gate, and qubit cost scalings of various second-quantized fermion-to-qubit encodings to perform the time evolution of a single fermionic term (eg. $e^{-i\theta(a_p^\dagger a_q + a_q^\dagger a_p)}$).

Encoding	Gate Costs	Qubit Cost
Sorted-list [17]	$\mathcal{O}(N \log M)$ Clifford $\mathcal{O}(N \log M)$ T gate	$\mathcal{O}(N \log M)$
Jordan-Wigner & Parity [11]	$\mathcal{O}(M)$ Clifford $\mathcal{O}(1)$ T gate	$\mathcal{O}(M)$
Bravyi-Kitaev [26]	$\mathcal{O}(\log M)$ Clifford $\mathcal{O}(1)$ T gate	$\mathcal{O}(M)$
Qubit Efficient [19] & Random Linear [20]	$\mathcal{O}(M^N)$ Clifford $\mathcal{O}(M^N)$ T gate	$\mathcal{O}(N \log M)$
Segment [21]	$\mathcal{O}(N^2)$ Clifford $\mathcal{O}(N^2)$ T gate	$\mathcal{O}(M - M/N)$
Polylog [22]	$\mathcal{O}(N^2 \log^5 M)$ Clifford $\mathcal{O}(N^2 \log^5 M)$ T gate	$\mathcal{O}(N^2 \log^4 M)$

It is important to emphasize that L and λ scales similarly between the two encodings scheme. For the Jordan-Wigner encoding, the fermionic operators are decomposed into Pauli strings, yielding $L = \mathcal{O}(M^4)$. While theoretically possible, the decomposition to Pauli strings is not be feasible for the sorted-list encoding due to an exponential blow-up to $L = \mathcal{O}(M^N)$ [21]. Instead, circuits implementing the time evolution of Hermitian fermionic operators are directly constructed as detailed in Section A.3 of the SI. This approach eliminates the need for decomposition, requiring only the grouping of conjugate operators to ensure Hermiticity (e.g. group $a_p^\dagger a_q$ with $a_q^\dagger a_p$ to form $a_p^\dagger a_q + a_q^\dagger a_p$ and $a_p^\dagger a_q - i a_q^\dagger a_p$). Consequently, L also scales as $\mathcal{O}(M^4)$ for the sorted-list encoding. The use of Hermitian fermionic operators for the sorted-list encoding also results in similar λ scalings between the two,

typically varying between $\mathcal{O}(M)$ to $\mathcal{O}(M^3)$ depending on the specific system [8].

Trotterization for the plane-wave basis. Unlike the MO basis where the h_{pq} and h_{pqrs} coefficients are arbitrary, the plane-wave (along with its dual basis) offers a more structured Hamiltonian, albeit at the trade-off of requiring a much larger basis for equivalent precision. This allows us to formulate a tighter bound for the gate cost of performing the RPE algorithm, demonstrated in Theorem 2.

Theorem 2. *The ground-state energy of the Hamiltonian H in the plane-wave basis can be calculated to precision ε using the RPE algorithm, where the gate costs (both Clifford and T-gates) scale as:*

$$\mathcal{O} \left(\left(\frac{M^{8/3}}{N^{2/3}} + M^{7/3} N^{2/3} \right) \frac{5^{p/2-1} (M^{1/3} N^{2/3})^{1/p}}{\varepsilon^{1+1/p}} \right)$$

for the second-quantized Jordan-Wigner encoding, and

$$\tilde{\mathcal{O}} \left(\left(M^{2/3} N^{4/3} + M^{1/3} N^{8/3} \right) \frac{5^{p/2-1} (M^{1/3} N^{2/3})^{1/p}}{\varepsilon^{1+1/p}} \right)$$

for both the first-quantized encoding and the second-quantized sorted-list encoding.

Due to the sheer number of plane-waves required compared to the other parameters, M is the most crucial parameter determining the overall cost. As a result, both the first-quantized and sorted-list encodings are most efficient with a leading order gate cost of $\tilde{\mathcal{O}}(M^{2/3+1/p})$, compared to $\mathcal{O}(M^{8/3+1/p})$ for the Jordan-Wigner encoding. This demonstrates a significant advantage for the $\mathcal{O}(N \log M)$ qubit encodings in this basis.

The gate costs of Theorem 2 are also significantly lower compared to the MO basis. To explain this superior scaling, we partition the Hamiltonian into kinetic $\hat{T}$, external potential $\hat{U}$, and coulomb potential $\hat{V}$ terms: $H = \hat{T} + \hat{U} + \hat{V}$. The kinetic term $\hat{T}$ is diagonal in

the plane-wave basis, while the potential terms $\hat{U} + \hat{V}$ are diagonal in the plane-wave dual basis. Since diagonal terms are mutually commuting, $\hat{T}$ and $\hat{U} + \hat{V}$ can be implemented exactly without any intrinsic trotter errors. Consequently, we effectively have $L = 2$ for this Hamiltonian decomposition, represented by $\hat{T}$ and $\hat{U} + \hat{V}$. Due to this low value of L , the deterministic formula is evidently more efficient for this Hamiltonian.

For the Jordan-Wigner encoding, both $\hat{T}$ and $\hat{U} + \hat{V}$ are encoded to Pauli strings of $\mathcal{O}(1)$ weight in their respective bases. The time evolution of these terms can be implemented with $\mathcal{O}(1)$ Clifford and one $R_Z(\theta)$ gate each. On the other hand, efficient construction of the first quantized Trotter circuit relies on phase kickback [14] to implement the diagonal gates. A drawback of the this approach is that, unlike the Jordan-Wigner encodings where the Hamiltonian coefficients are calculated classically and incorporated into the rotation angles of the $R_Z(\theta)$ gates, these coefficients needs to be calculated in the quantum circuit for the first-quantized approach, implying higher constant factors despite the favourable scaling.

While the sorted-list encoding implementation can be adapted from either approach, the first-quantized approach is chosen for Theorem 2. This is because, unlike the massive gate cost reductions observed in the Jordan-Wigner encoding, the sorted-list encoding still maintains the $\mathcal{O}(N \log M)$ Clifford and T gate scaling when implementing $\hat{T}$ and $\hat{U} + \hat{V}$ operators. Conversely, converting the first-quantized implementation to the sorted-list encoding is more straightforward, as the action of $\hat{T}$ and $\hat{U} + \hat{V}$ acts identically between the first-quantized and sorted-list encodings, resulting in identical scaling between the two.

Costings on real-world systems. In this section, we provide numerical results for the T gate and qubit costs of performing RPE calculations for several real-world molecular systems. We chose to highlight the second-order ($p = 2$) Suzuki Trotter, qDRIFT, and the hybrid formula using both the Jordan-Wigner and the sorted-list encoding, following the RPE cost analysis done in a previous work [2]. The qubit and T gate costs are shown in Table III.

In terms of the qubit count, it is evident that the sorted-list encoding shows an advantage in the $N \ll M$ regime, with the break-even point observed around $N/M \approx 0.1$. However, as previously discussed, the Jordan-Wigner encoding requires only the single-qubit rotation gates as its only non-Clifford gate, while the sorted-list encoding requires Toffoli gates, in addition to the single-qubit rotation gates, due to its inherent binary comparison operations. Consequently, the sorted-list encoding typically requires 2-4 orders of magnitude more T-gates for implementation. This highlights a clear trade-off between qubit efficiency and T-gate complexity across different encoding schemes and Trotterization strategies.

The major advantage of the above trotterization cir-

cuit is its conceptual simplicity as the Hamiltonian simulation can implemented as a sequence of time evolution circuits, making it a more realistic choice for early fault-tolerant implementations. With the exception of the first-quantized and sorted-list implementations in the plane-wave basis, the exponentiation of the Hermitian terms turns the h_{pq} and h_{pqrs} coefficients of the Hamiltonian into rotation angles applied on $R_Z(\theta)$ gates. In comparison, qubitization circuits uses QROMs to load those Hamiltonian coefficients into the circuit, a significantly more complex operation.

The complexity of the qubitization circuit trades off with more efficient scaling with respect to the precision parameter ε and Hamiltonian 1-norm λ . In terms of the precision parameter, the QPE and RPE algorithms both contribute a $1/\varepsilon$ factor towards the gate cost for phase estimation accuracy. Moreover, the trotter product formula itself contributes an additional $1/\varepsilon^{1/p}$ for the deterministic formula and $1/\varepsilon$ for the random formula. This results in the overall ε scaling found in Theorem 1 and 2. On the other hand, qubitization offers a logarithmic dependence on the error scaling for the simulation primitive, resulting in an overall gate cost scaling of $\mathcal{O}(1/\varepsilon)$ when combined with the QPE algorithm. Qubitization also offers similar advantages over the Hamiltonian 1-norm λ , where the gate cost has a linear dependence with λ , compared to the quadratic dependence for the qDRIFT trotter formula.

Qubitization

For qubitization circuits, the Hamiltonian adopts a linear combination of unitaries (LCU) representation:

$$H = \sum_{l=0}^{L-1} h_l U_l,$$

where h_l are positive real coefficients ($h_l \in \mathbb{R}^+$), and U_l are unitary operators. The QPE algorithm is then implemented via the Szegedy quantum walk [18]

$$\mathcal{W} = (2 \text{PREPARE} |0\rangle\langle 0| \text{PREPARE}^\dagger \otimes \mathbb{I} - \mathbb{I}) \text{SELECT},$$

where PREPARE and SELECT constitute the major gate and qubit costs of the algorithm. These circuits can be described at a high-level as:

$$\begin{aligned} \text{PREPARE} |0\rangle &= \sum_{l=0}^{L-1} \sqrt{\frac{|h_l|}{\lambda}} |l\rangle, \\ \text{SELECT} |l\rangle |\psi\rangle &= |l\rangle U_l |\psi\rangle, \end{aligned}$$

with $\lambda = \sum_{l=0}^{L-1} |h_l|$ being the 1-norm of the Hamiltonian. To obtain the ground-state energy of the Hamiltonian with precision ε , the QPE algorithm requires $\mathcal{O}(\lambda/\varepsilon)$ applications of $\mathcal{W}$.

Qubitization in the MO basis Unlike the trotterization case where the Hamiltonian coefficients h_{pq} and h_{pqrs} can be arbitrary, we restrict our Hamiltonian such

Table III: Qubit count and T gate cost of performing an RPE calculation using the deterministic, random, and partially random trotter formulas [2] of molecular systems using MO orbitals. Due to the high classical costs in determining the partition point between deterministic and random parts, we skipped the cost analysis of the larger systems. Qubit counts where the sorted-list encoding has an advantage over the Jordan-Wigner encoding are shown in bold.

System	Orbitals	Jordan-Wigner				Sorted-list			
		Qubits	Suzuki	qDRIFT	Partial	Qubit	Suzuki	qDRIFT	Partial
H ₂ O	7 (STO-3G)	16	2×10^9	4×10^{11}	5×10^9	55	7×10^{11}	7×10^{14}	7×10^{11}
10 elec	19 (6-311G)	40	9×10^{11}	2×10^{13}	1×10^{12}	77	4×10^{14}	4×10^{16}	4×10^{14}
Model	24 (cc-pvdz)	50	3×10^{12}	3×10^{13}	4×10^{12}	77	1×10^{15}	6×10^{16}	1×10^{15}
System	58 (cc-pvtz)	118	1×10^{15}	4×10^{15}	1×10^{15}	88	5×10^{17}	5×10^{18}	4×10^{17}
	115 (cc-pvqz)	232	1×10^{17}	2×10^{17}	1×10^{17}	99	6×10^{19}	3×10^{20}	4×10^{19}
	201 (cc-pv5z)	404	7×10^{18}	4×10^{18}		110	3×10^{21}	7×10^{21}	
CO ₂	15 (STO-3G)	32	1×10^{11}	5×10^{12}	2×10^{11}	126	1×10^{14}	3×10^{16}	1×10^{14}
22 elec	39 (6-311G)	80	6×10^{13}	3×10^{14}	5×10^{13}	172	7×10^{16}	2×10^{18}	4×10^{16}
Model	42 (cc-pvdz)	86	7×10^{13}	3×10^{14}	7×10^{13}	172	8×10^{16}	2×10^{18}	6×10^{16}
System	90 (cc-pvtz)	182	1×10^{16}	2×10^{16}	9×10^{15}	195	1×10^{19}	9×10^{19}	9×10^{18}
	165 (cc-pvqz)	332	8×10^{17}	5×10^{17}		218	9×10^{20}	2×10^{21}	
	273 (cc-pv5z)	548	2×10^{19}	8×10^{18}		241	3×10^{22}	3×10^{22}	
EC	34 (STO-3G)	70	4×10^{14}	6×10^{14}	4×10^{13}	340	2×10^{17}	2×10^{18}	6×10^{16}
46 elec	90 (6-311G)	182	2×10^{17}	9×10^{16}		387	1×10^{20}	2×10^{20}	
Battery	104 (cc-pvdz)	210	3×10^{17}	8×10^{16}		387	4×10^{20}	5×10^{20}	
Materials	236 (cc-pvtz)	474	7×10^{19}	9×10^{18}		434	1×10^{23}	7×10^{22}	
	460 (cc-pvqz)	922	5×10^{21}	3×10^{20}		481	1×10^{25}	4×10^{24}	
LiPF ₆	44 (STO-3G)	90	5×10^{13}	1×10^{14}	2×10^{14}	522	2×10^{18}	1×10^{19}	5×10^{17}
72 elec	112 (6-311G)	226	3×10^{16}	2×10^{16}		595	1×10^{21}	1×10^{21}	
Battery	116 (cc-pvdz)	234	1×10^{17}	5×10^{16}		595	1×10^{21}	1×10^{21}	
Materials	244 (cc-pvtz)	490	4×10^{19}	8×10^{18}		668	3×10^{23}	1×10^{23}	
	444 (cc-pvqz)	890	4×10^{21}	3×10^{20}		741	2×10^{25}	5×10^{24}	

that $h_{pq} \in \mathbb{R}$ and $h_{pqrs} \in \mathbb{R}$. Furthermore, we assume that the h_{pqrs} coefficients adopt the 8-fold symmetry, which is typically obeyed by the Hamiltonians of molecular systems. In this case, the gate and qubit costs of the QPE algorithm for such Hamiltonians are formalized in Table IV.

A natural unitary decomposition of the Hamiltonian here involves the Majorana operators:

$$\gamma_{p,\alpha,0} = a_{p,\alpha}^\dagger + a_{p,\alpha} \quad \gamma_{p,\alpha,1} = ia_{p,\alpha}^\dagger - ia_{p,\alpha},$$

with the SELECT circuit implementing a product of either two or four of them for the one- and two-electron terms, respectively. The Jordan-Wigner implementation of this involves unitary iteration circuits [18] requiring $\mathcal{O}(M)$ gates. In contrast, the sorted-list implementation, detailed in Section A.4 of the SI, has a gate cost scaling of $\mathcal{O}(N \log M)$. This suggests a clear theoretical advantage for the sorted-list encoding over the Jordan-Wigner in the $N \ll M$ limit.

The implementation of the PREPARE circuit, however, remains the same between the two encodings. To mitigate the $L = \mathcal{O}(M^4)$ scaling, various rank-reduction techniques have been proposed to reduce its qubit and gate costs, as listed in Table IV, typically at the cost of expensive classical preprocessing of the Hamiltonian. De-

spite these efforts, the gate cost scaling of the PREPARE circuit still dominates over the SELECT circuit. This result in similar overall gate cost scalings between the two encodings when integrated together into the QPE algorithm. A notable exception is the tensor hypercontraction approach [8], where a simplified PREPARE circuit necessitates a more complex SELECT circuit, for which the Jordan-Wigner encoding fares better than the sorted-list encoding.

Qubitization for the plane-wave basis The highly structured nature of the plane-wave basis set also significantly benefits the implementation of the qubitization circuits. We adapted a highly efficient first-quantized implementation of the Szegedy quantum walk operator [16] for our implementation of the QPE algorithm for plane-wave Hamiltonians with the sorted-list encoding. The gate costs are formalized in Theorem 3.

Theorem 3. *The ground-state energy of a Hamiltonian in the plane-wave basis can be calculated to precision ε using the QPE algorithm with the second-quantized sorted-list encoding using*

$$\tilde{\mathcal{O}}([N^{1/3}M^{5/3} + N^{5/3}M^{4/3} + N^{2/3}M^{7/3}]/\varepsilon)$$

gates and $\tilde{\mathcal{O}}(N)$ qubits.

Table IV: Clifford gate, T gate, and qubit cost scaling for obtaining the ground-state energy of a Hamiltonian of 1-norm λ with precision ε using the QPE algorithm. Various rank reduction techniques to mitigate the $\mathcal{O}(M^4)$ complexity of the two-electron terms in the MO basis are included. Due to the use of Quantum Read-Only Memories (QROMs) in the QPE algorithm, we provided two QROM implementations [27], one that minimizes ancilla qubits (“Minimize ancilla”) and another that minimize the T gate cost (“Minimize T gate”).

Method	Minimize Ancilla		Minimize T gate		Resources
	Jordan-Wigner	Sorted-List	Jordan-Wigner	Sorted-List	
Sparse [6]	$\tilde{\mathcal{O}}(\lambda M^4/\varepsilon)$		$\tilde{\mathcal{O}}(\lambda M^4/\varepsilon)$		Cliffords
	$\tilde{\mathcal{O}}(\lambda M^4/\varepsilon)$		$\tilde{\mathcal{O}}(\lambda M^2/\varepsilon)$		T gates
	$\tilde{\mathcal{O}}(M)$	$\tilde{\mathcal{O}}(N)$	$\tilde{\mathcal{O}}(M^2)$		Qubits
Single-Factorization [6]	$\tilde{\mathcal{O}}(\lambda M^3/\varepsilon)$		$\tilde{\mathcal{O}}(\lambda M^3/\varepsilon)$		Cliffords
	$\tilde{\mathcal{O}}(\lambda M^3/\varepsilon)$		$\tilde{\mathcal{O}}(\lambda M^{3/2}/\varepsilon)$		T gates
	$\tilde{\mathcal{O}}(M)$	$\tilde{\mathcal{O}}(N)$	$\tilde{\mathcal{O}}(M^{3/2})$		Qubits
Double-Factorization [7]	$\tilde{\mathcal{O}}(\lambda M^3/\varepsilon)$		$\tilde{\mathcal{O}}(\lambda M^3/\varepsilon)$		Cliffords
	$\tilde{\mathcal{O}}(\lambda M^3/\varepsilon)$		$\tilde{\mathcal{O}}(\lambda M^{3/2}/\varepsilon)$		T gates
	$\tilde{\mathcal{O}}(M)$	$\tilde{\mathcal{O}}(N)$	$\tilde{\mathcal{O}}(M^{3/2})$		Qubits
Tensor Hypercontraction [8]	$\tilde{\mathcal{O}}(\lambda M^2/\varepsilon)$		$\tilde{\mathcal{O}}(\lambda M^2/\varepsilon)$		Cliffords
	$\tilde{\mathcal{O}}(\lambda M^2/\varepsilon)$		$\tilde{\mathcal{O}}(\lambda M/\varepsilon)$	$\tilde{\mathcal{O}}(\lambda M N/\varepsilon)$	T gates
	$\tilde{\mathcal{O}}(M)$	$\tilde{\mathcal{O}}(N)$	$\tilde{\mathcal{O}}(M)$		Qubits

Unfortunately, the original first-quantized QPE algorithm is significantly more efficient, requiring $\tilde{\mathcal{O}}([N^{4/3}M^{2/3} + N^{8/3}M^{1/3}]/\varepsilon)$ gates and $\tilde{\mathcal{O}}(N)$ qubits [16]. Unlike trotterization, where parity in cost scaling is maintained between the first-quantized and the sorted-list encodings, the sorted-list qubitization circuit exhibit worse gate complexity due to the algorithm’s dependence on the Hamiltonian 1-norm λ . For plane-wave Hamiltonians, the first-quantized λ scales as $\mathcal{O}(N^{-1/3}M^{2/3} + N^{5/3}M^{1/3})$, compared to $\mathcal{O}(N^{-2/3}M^{5/3} + N^{2/3}M^{4/3} + N^{-1/3}M^{7/3})$ for the second-quantized case. Table V shows these λ difference decomposed into the kinetic, external potential, and coulomb potential terms. As a result, despite similar gate cost scalings per application of the walk operator, the first-quantized approach yields a more efficient overall QPE circuit due to its superior Hamiltonian 1-norm complexity scaling, which directly translates to fewer applications of the walk operator.

Costings of real-world systems. Here, we present numerical results for the T gate and qubit costs of QPE calculations for several real-world molecular systems. For our cost analysis in the MO basis, we utilized the single-factorization form of the Hamiltonian [6], augmented with additional circuit optimizations suggested by another work [8]. We opted not to highlight the sorted-list implementations of the double-factorization [7] and tensor hypercontraction [8] methods as the $\mathcal{O}(M)$ sequence of controlled Pauli rotations in their implementations (Section VII.C.2 of the Supporting Information of a previous paper [7]) would translate to $\mathcal{O}(M)$ Majorana operations. This would result in a less favourable T gate scaling of $\mathcal{O}(MN \log M)$ as compared to the $\mathcal{O}(M)$

Table V: Scaling of λ between first-quantized and second-quantized form of the Hamiltonian using the plane-wave basis set [16]. In total, we have

$$\lambda = \lambda_T + \lambda_U + \lambda_V. \text{ Thus,}$$

$$\lambda = \mathcal{O}(N^{-1/3}M^{2/3} + N^{5/3}M^{1/3}) \text{ for the first-quantized Hamiltonian and}$$

$$\lambda = \mathcal{O}(N^{-2/3}M^{5/3} + N^{2/3}M^{4/3} + N^{-1/3}M^{7/3}) \text{ for the second-quantized Hamiltonian.}$$

Terms	First-quantized	Second-quantized
λ_T (kinetic)	$\mathcal{O}(N^{-1/3}M^{2/3})$	$\mathcal{O}(N^{-2/3}M^{5/3})$
λ_U (external potential)	$\mathcal{O}(N^{5/3}M^{1/3})$	$\mathcal{O}(N^{2/3}M^{4/3})$
λ_V (two-electron)	$\mathcal{O}(N^{5/3}M^{1/3})$	$\mathcal{O}(N^{-1/3}M^{7/3})$

scaling achieved with the Jordan-Wigner representation.

Table VI presents the calculated qubit and T gate costs required for QPE on several model chemical systems with results from previous works for comparison. Notably, the T gate costs in the table are reported as Toffoli counts from the original works and will be converted to T gate counts for our analysis, with 2 T gates for every Toffoli gates [28]. Given that the T gate cost of the QROM can potentially be reduced through the use of either “dirty” or “clean” qubits [27], we included a cost analysis for two QROM implementations: one that minimizes the ancilla qubits and another that minimizes the T gate cost. Our comparison with the Jordan-Wigner encoding demonstrates that the sorted-list encoding maintains similar T gate costs, even for FeMoCo systems where the number

of electrons is greater than or equal to half the number of spin orbitals. For systems with a lower electron filling ratio, such as the EC and LiPF_6 systems, we achieved comparable T gate costs to the Jordan-Wigner encoding but with significantly reduced qubit requirements.

We also performed cost analysis on the same systems using the plane-wave basis. As mentioned in the previously, the inherent efficient scaling of the first-quantized algorithm was not preserved when adapted to the sorted-list encoding. This discrepancy is attributed to the dependence of the qubitization circuits on the Hamiltonian 1-norm λ . Table VII presents a cost comparison between the first-quantized implementation [16] and our sorted-list implementation.

Discussion

In this paper, we presented the cost analysis of phase estimation algorithms via trotterization and qubitization utilizing the sorted-list encoding [17]. The sorted-list encoding demonstrates a $\mathcal{O}(N \log M)$ scaling with respect to both the qubit count and the gate cost to implement a fermionic operator, offering a qubit advantage over conventional $\mathcal{O}(M)$ qubit encodings in the $N \ll M$ regime. While this is an improvement over previous electron-conserving encoding schemes, its trade-offs in the context of phase estimation algorithms had not yet been thoroughly analyzed. We also obtained numerical estimates for the qubit and T gate costs of phase estimation for several chemical systems, including H_2O , CO_2 , Ethylene Carbonate (EC), and LiPF_6 .

For the trotterization circuits in the MO basis, the sorted-list encoding achieved an obvious qubit advantage when $N \ll M$, with numerical experiments showing the break-even point at $N/M \approx 0.1$. This, however, comes at a cost of an increased T gate cost compared to the Jordan-Wigner encoding, at roughly 2-4 orders of magnitude higher for the chemical systems we tested. Despite this, our theoretical analysis of the sorted-list encoding predicted an eventual quantum resource advantage over the Jordan-Wigner encoding in the $N/M \rightarrow 0$ limit, primarily due to better Clifford gate scaling. In contrast, the structured nature of the plane-wave basis allowed for a different, more efficient implementation compared to the MO basis. Consequently, the sorted-list encoding offers both qubit and gate cost advantage for trotterization circuits in the plane-wave basis. Due to the similarity between the first-quantized and the sorted-list encoding, we are able to leverage the hybrid quantization scheme [32] to obtain similar gate cost scalings between the two.

Regarding qubitization circuits, we obtained similar gate and qubit costs between the Jordan-Wigner and sorted-list encodings for calculations in the MO basis. This parity arises because the gate and qubit complexities are dominated by the PREPARE circuit, which is identical for both encodings. Numerical estimation of the qubitization circuits further demonstrated these comparable gate costs. Unfortunately, less favourable results were obtained when using the plane-wave basis. Here, we found that our sorted-list implementation exhibited

worse gate scaling compared to the first-quantized encoding, a disparity attributed to the much lower Hamiltonian 1-norm associated with the first-quantized Hamiltonian. While the T gate costs per step were similar between the first-quantized circuit and our second-quantized sorted-list implementation, the latter required a greater number of QPE steps to obtain equivalent precision. This disparity resulted in the first-quantized encoding achieving T gate costs that were 5 to 10 orders of magnitude lower, depending on system size.

When performing phase estimation on large molecules, the sublinear gate scaling of the plane-wave qubitization circuits [16] is very attractive and represents the best known scaling to date. This is complemented with an $\tilde{\mathcal{O}}(N)$ qubit scaling, which is ideal for the large M inherent to the plane-wave basis. However, utilizing the plane-wave basis for molecular systems presents several caveats. The most prominent is the necessity to eliminate interactions between periodic images. Such errors can typically be reduced by increasing the simulation cell size, which incurs larger M , or by modifying the Coulomb potential to ignore long-range interactions [33–35], which typically increases the complexity of the PREPARE circuits.

Conversely, for the simulation of smaller systems in NISQ or near-term partially fault-tolerant devices, the QROMs required for qubitization circuits presents implementation challenges in achieving sufficiently low error rates [36]. As a result, trotterization circuits in the MO basis using the Jordan-Wigner encoding may be preferable in this scenario. The larger scaling is offset by the significantly smaller M required for these molecules.

Future work will focus on further optimizing the trotterization and qubitization circuits to reduce their constant overheads. For qubitization circuits using the MO basis, there is an clear opportunity to further exploit the particle conservation symmetry inherent in chemical Hamiltonians. Techniques such as the Block-Invariant Symmetry Shift (BLISS) [37] could be particularly beneficial here, as they can reduce the 1-norm of Hamiltonians and, consequently, the T gate cost of the qubitization circuits. Additionally, the applicability of other symmetries, such as S_z symmetry and molecular point group symmetries, within the context of the sorted-list encoding warrants further investigation. Finally, the efficient implementation of qubitization circuits for double-factorized and tensor hypercontracted Hamiltonians in the sorted-list encoding remains an open research question.

Methods

Proof of Theorem 1. For the sorted-list encoding, we decompose the Hamiltonian into a linear combination of the following hermitian operators:

$$a_p^\dagger a_q + a_q^\dagger a_p, \quad i a_p^\dagger a_q - i a_q^\dagger a_p, \\ a_p^\dagger a_q^\dagger a_r a_s + a_s^\dagger a_r^\dagger a_q a_p, \quad \text{and} \quad i a_p^\dagger a_q^\dagger a_r a_s - i a_s^\dagger a_r^\dagger a_q a_p.$$

Table VI: Comparison of the qubitization cost with previous works. For the FeMoCo system, we compare our results with the single-factorized Hamiltonian [6] using the Jordan-Wigner encoding. For the EC and LiPF₆ systems, we compare our results with the double-factorized Hamiltonian[29] using the Jordan-Wigner encoding. Toffoli costs of the previous work are converted to T gates with 2 T gates for every Toffoli gates [28].

Type	Spatial Orbitals	Previous Work				Our Work			
		Minimize Qubits		Minimize T gate		Minimize Qubits		Minimize T gate	
		Qubits	T gate	Qubits	T gate	Qubits	T gate	Qubits	T gate
FeMoCo									
RWSWT [30]	54	378	4.2×10^{13}	3024	2.4×10^{12}	703	1.4×10^{11}	835	1.2×10^{11}
LLDUC [31]	76	437	4.0×10^{13}	3143	2.0×10^{12}	1349	1.8×10^{11}	1368	1.6×10^{11}
EC									
STO-3G	34			2685	6.4×10^{10}	622	5.5×10^{10}	627	4.8×10^{10}
6-311G	90			14492	1.7×10^{12}	707	1.6×10^{12}	892	1.2×10^{12}
cc-pvdz	104			16698	2.6×10^{12}	712	3.1×10^{12}	1232	2.2×10^{12}
cc-pvtz	236			81958	6.2×10^{13}	791	1.1×10^{14}	2154	5.9×10^{13}
LiPF ₆									
STO-3G	44			3507	2.0×10^{11}	863	1.9×10^{11}	847	1.7×10^{11}
6-311G	112			18423	3.8×10^{12}	972	4.6×10^{12}	1440	3.5×10^{12}
cc-pvdz	116			18600	3.4×10^{12}	972	4.6×10^{12}	1440	3.5×10^{12}
cc-pvtz	244			84721	6.6×10^{13}	1076	1.0×10^{14}	2388	6.7×10^{13}

Table VII: Costings of the qubitization operators for molecular systems with the plane-wave basis set. We added the first-quantized plane-wave implementation [16] for comparison.

System	PWs	First-Quantized (Previous work[16])				Second-Quantized (Our work)			
		nQPE	Qubits	T gate/Step	T gate Total	nQPE	Qubits	T gate/Step	T gate Total
H ₂ O	4.1×10^3	20	825	3.1×10^3	3.2×10^9	36	1005	1.2×10^4	8.3×10^{14}
10 elec	3.3×10^4	21	1015	3.7×10^3	7.8×10^9	43	1237	1.5×10^4	1.3×10^{17}
	2.6×10^5	22	1219	4.5×10^3	1.9×10^{10}	50	1483	1.7×10^4	2.0×10^{19}
	2.1×10^6	24	1440	5.2×10^3	8.7×10^{10}	57	1743	2.0×10^4	2.9×10^{21}
CO ₂	4.1×10^3	22	1021	4.5×10^3	1.9×10^{10}	36	1229	2.0×10^4	1.4×10^{15}
22 elec	3.3×10^4	23	1255	5.5×10^3	4.6×10^{10}	43	1505	2.5×10^4	2.2×10^{17}
	2.6×10^5	25	1506	6.5×10^3	2.2×10^{11}	50	1795	2.9×10^4	3.3×10^{19}
	2.1×10^6	26	1768	7.5×10^3	5.1×10^{11}	57	2099	3.4×10^4	4.8×10^{21}
EC	4.1×10^3	25	1363	7.2×10^3	2.4×10^{11}	36	1632	3.7×10^4	2.5×10^{15}
46 elec	3.3×10^4	26	1677	8.7×10^3	5.9×10^{11}	43	1988	4.4×10^4	3.9×10^{17}
	2.6×10^5	27	2005	1.0×10^4	1.4×10^{12}	50	2358	5.2×10^4	5.9×10^{19}
	2.1×10^6	28	2347	1.2×10^4	3.2×10^{12}	57	2742	6.0×10^4	8.7×10^{21}
LiPF ₆	4.1×10^3	26	1701	1.0×10^4	6.7×10^{11}	36	2043	5.4×10^4	3.7×10^{15}
72 elec	3.3×10^4	27	2097	1.2×10^4	1.6×10^{12}	43	2481	6.6×10^4	5.8×10^{17}
	2.6×10^5	28	2507	1.4×10^4	3.9×10^{12}	50	2933	7.7×10^4	8.7×10^{19}
	2.1×10^6	29	2931	1.7×10^4	9.0×10^{12}	57	3399	8.9×10^4	1.3×10^{22}

Thus, we must construct circuit implementations for the time evolution of the above unitaries:

$$e^{i\theta(a_p^\dagger a_q + a_q^\dagger a_p)}, \quad e^{i\theta(ia_p^\dagger a_q - ia_q^\dagger a_p)}, \quad (1)$$

$$e^{i\theta(a_p^\dagger a_q^\dagger a_r a_s + a_s^\dagger a_r^\dagger a_q a_p)}, \quad \text{and} \quad e^{i\theta(ia_p^\dagger a_q^\dagger a_r a_s - ia_s^\dagger a_r^\dagger a_q a_p)}.$$

We detail these circuit implementations in Section A.3 of the SI, with exact gate costs shown in Table A.6 of the SI. When decomposed to a Clifford+T universal gate set,

each of these unitaries in Equation 1 requires $\mathcal{O}(N \log M)$ Clifford gates, T gates and qubits for its implementation.

Next, Table A.2 of the SI specifies the total number of applications of the unitaries from Equation 1 needed for the RPE algorithm to calculate the ground-state energy with precision ε . By multiplying the costs from the table by the $\mathcal{O}(N \log M)$ gate costs for each unitary application, we reproduce the overall costs reported in the main text. $\square$

Proof of Theorem 2. To obtain the ground-state with precision ε , the RPE algorithm [2] requires $\mathcal{O}(1/\varepsilon\delta)$ applications of the p^{th} -order trotter formula $S_p(\delta)$, which approximates $e^{-iH\delta}$. The time step δ is chosen such that the trotter error is bounded by $\varepsilon\delta$. Previous work has established tight bounds for such Trotter errors [38], which derived:

$$|S_p(\delta) - e^{-iH\delta}| = \mathcal{O}\left(\left(\frac{M^{2/3}}{N^{2/3}} + M^{1/3}N^{2/3}\right)^p M^{1/3}N^{2/3}\delta^{p+1}\right).$$

To bound this error below $\varepsilon\delta$, we find the required δ :

$$\delta \sim \mathcal{O}\left(\varepsilon^{1/p} \left(\frac{M^{2/3}}{N^{2/3}} + M^{1/3}N^{2/3}\right)^{-1} (M^{1/3}N^{2/3})^{-1/p}\right).$$

substituting this δ into $\mathcal{O}(1/\varepsilon\delta)$ yields the total number of $S_p(\delta)$ applications:

$$\mathcal{O}\left(\frac{1}{\varepsilon\delta}\right) = \mathcal{O}\left(\left(\frac{M^{2/3}}{N^{2/3}} + M^{1/3}N^{2/3}\right) \frac{(M^{1/3}N^{2/3})^{1/p}}{\varepsilon^{1+1/p}}\right). \quad (2)$$

With the required number of $S_p(\delta)$ applications determined, we are left with obtaining the gate cost to implement $S_p(\delta)$ for each of the encodings. Unlike the MO basis, we exclusively use the deterministic trotter formula as the plane-wave Hamiltonian can be decomposed into $L = 2$ terms, represented by the kinetic $\hat{T}$ and potential $\hat{U} + \hat{V}$ terms. This decomposition is discussed in Section A.2.1.1 of the SI.

Following a previous Jordan-Wigner implementation [39], we find a gate cost of $\mathcal{O}(5^{p/2-1}M^2)$ for one application of $S_p(\delta)$. We included the $5^{p/2-1}$ factor from Equation A.2 of the SI for our cost analysis. For the first-quantized encoding, one application of $S_p(\delta)$ has a gate cost of $\tilde{\mathcal{O}}(5^{p/2-1}N^2)$ [14]. The gate costs in the main text for both encodings are then reproduced by multiplying these encoding specific costs for one $S_p(\delta)$ application by Equation 2.

For the cost analysis of the sorted-list encoding, we can adapt either the Jordan-Wigner or the first-quantized approaches. When adapting from the Jordan-Wigner implementation, each application of $S_p(\delta)$ requires $\mathcal{O}(5^{p/2-1}M^2)$ applications of $e^{i\theta a_p^\dagger a_p}$. Each of these applications contributes a major cost of $\mathcal{O}(N \log M)$ Clifford and T gates (see Table A.6 of the SI). Consequently, the Clifford and T gate costs of $S_p(\delta)$ scale as $\mathcal{O}(5^{p/2-1}M^2N \log M)$, which is worse than the original Jordan-Wigner encoding.

Conversely, using the first-quantized approach yields identical gate cost scaling. Consider the Slater determinant $|\nu_1\rangle \otimes \dots \otimes |\nu_N\rangle$ in the sorted-list encoding, where $\nu_1, \dots, \nu_N$ are indices of the occupied orbitals. The time evolution of the kinetic term $\hat{T}^{(2)}$ (see Section A.2.1.1 of

the SI) can be written as:

$$\begin{aligned} e^{-i\hat{T}^{(2)}t} |\nu_1\rangle \otimes \dots \otimes |\nu_N\rangle &= e^{-it \sum_{\nu=1}^M T_{\nu} a_{\nu}^\dagger a_{\nu}} |\nu_1\rangle \otimes \dots \otimes |\nu_N\rangle \\ &= \prod_{\nu=1}^M e^{-it T_{\nu} a_{\nu}^\dagger a_{\nu}} |\nu_1\rangle \otimes \dots \otimes |\nu_N\rangle \\ &= \prod_{i=1}^N e^{-iT_{\nu_i}t} |\nu_1\rangle \otimes \dots \otimes |\nu_N\rangle \\ &= e^{-i\hat{T}^{(1)}t} |\nu_1\rangle \otimes \dots \otimes |\nu_N\rangle. \end{aligned}$$

This shows that $e^{-i\hat{T}^{(2)}t} = e^{-i\hat{T}^{(1)}t}$ when applied to a qubit wavefunction in the sorted-list encoding. Therefore, we can use the same first-quantized implementation for the sorted-list encoding. Through similar arguments, the same can be said for the external $e^{-i\hat{U}^{(2)}t}$ and coulomb potential $e^{-i\hat{V}^{(2)}t}$ terms, maintaining the same cost for $S_p(\delta)$ between the first-quantized and sorted-list encoding. We are left with the implementation of the fermionic fast fourier transform circuits required to switch between the plane-wave for $\hat{T}^{(2)}$ and its dual basis for $\hat{U}^{(2)} + \hat{V}^{(2)}$. This can be done efficiently using the hybrid quantization scheme introduced in a previous work [32], where we convert the wavefunction to the first-quantized encoding, performed the QFT in the first-quantization, and converting back to the sorted-list encoding with a gate cost of $\tilde{\mathcal{O}}(N)$. This results in identical gate cost scaling for the entire RPE algorithm, as shown in the main text. $\square$

Proof of Theorem 3. We adapt the construction of the PREPARE and SELECT circuits for the sorted-list encoding from a first-quantized implementation [16]. The first-quantized implementation is summarized in Section A.2.2.2 of the SI while our adaptation is detailed in Section A.4.3 of the SI. For the sorted-list encoding, the PREPARE circuit costs $\tilde{\mathcal{O}}(1)$ gates while the SELECT circuit costs $\tilde{\mathcal{O}}(N)$. To obtain the ground-state with precision ε , the QPE circuit requires $\mathcal{O}(\lambda/\varepsilon)$ applications of PREPARE and SELECT circuits. substituting $\lambda = \mathcal{O}(N^{-2/3}M^{5/3} + N^{2/3}M^{4/3} + N^{-1/3}M^{7/3})$ from Table 5 of the main manuscript, we derive the total gate cost:

$$\tilde{\mathcal{O}}\left(\frac{\lambda N}{\varepsilon}\right) = \tilde{\mathcal{O}}\left(\frac{N^{1/3}M^{5/3} + N^{5/3}M^{4/3} + N^{2/3}M^{7/3}}{\varepsilon}\right). \quad \square$$

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AUTHOR CONTRIBUTIONS

The project was conceived by C.K. and Y.C. Theoretical results were proved by C.K. in discussion with M.H. Numerical simulations and analysis were performed by C.K. in discussion with A.H. All authors contributed to

the write-up.

COMPETING INTERESTS

The authors declare no competing interests.

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Supporting Information for “Benchmarking Quantum Simulation Methods”

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Appendix A.1: Details on Encoding Schemes

This section introduces previous works on the first-quantized encoding and the sorted-list encoding. For the rest of the document, we will use the notation $\log M$ to represent $\lceil \log_2 M \rceil$.

A.1.1. First-Quantized Encoding

The first-quantized representation uses the tensor products of the single-particle wavefunctions $\psi_1, \dots, \psi_N$ as a basis, called Hartree products. As a single Hartree product does not inherently obey the Pauli exclusion principle, the anti-symmetry requirements for fermions must be explicitly expressed as a linear combination of Hartree products. The resulting antisymmetrized wavefunction can then be expressed as a Slater determinant, which is a linear combination of $N!$ Hartree products, expressed as a matrix determinant:

$$[\psi_1, \dots, \psi_N] = \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_1(\mathbf{x}_1) & \cdots & \psi_N(\mathbf{x}_1) \\ \vdots & \ddots & \vdots \\ \psi_1(\mathbf{x}_N) & \cdots & \psi_N(\mathbf{x}_N) \end{vmatrix}.$$

The Slater determinant on the left-hand side above are the basis for the second-quantized representation while the Hartree products are the basis for the first-quantized representation. The first-quantized encoding [14–16] encodes a single Hartree product as follows:

$$\begin{aligned} \Psi &= \psi_{i_1}(\mathbf{x}_1) \otimes \psi_{i_2}(\mathbf{x}_2) \otimes \cdots \otimes \psi_{i_N}(\mathbf{x}_N), \\ \mathcal{E}(\Psi) &= |i_1\rangle \otimes |i_2\rangle \otimes \cdots \otimes |i_N\rangle. \end{aligned}$$

Here, $|i_1\rangle, \dots, |i_N\rangle = |1\rangle, \dots, |M\rangle$ represents the binary representations of the indices to the single-particle basis states $|\psi_1(\mathbf{x})\rangle, \dots, |\psi_M(\mathbf{x})\rangle$. Each index $|i\rangle$ is encoded using $\lceil \log_2 M \rceil$ qubits, resulting in a total of $N \lceil \log_2 M \rceil$ qubits for N electrons. Unlike the sorted-list encoding, the indices are not sorted, as different orderings of $|i_1\rangle, \dots, |i_n\rangle$ correspond to distinct Hartree products.

The antisymmetrization of the electrons in this encoding must be handled explicitly during the quantum simulation process. Fortunately, the Hamiltonian evolution discussed in the main manuscript preserves this antisymmetric property, as long as the input wavefunction is antisymmetric. As a result, explicit antisymmetrization is only required during state initialization [15].

A.1.2. Sorted-List Encoding

The previous work [17] that proposed the sorted-list encoding introduced several variants of the optimal $\mathcal{O}(N \log M)$ qubit second-quantized encodings. We used the variant detailed in Section 4.2 of their paper, which they refer to as the sorted-list encoding. This variation has the simplest encoding along with the most efficient circuit implementations in terms of gate count. Other variations increase the complexity of the encoding and circuit implementation to save either the qubit count and/or the circuit depth. This variation also allows us to create a very compact circuit for the Trotterization implementation, along with efficient conversions between the first-quantized and second-quantized encodings.

Given M orbitals, let $\mathbf{x} \in \mathbb{F}_2^M$ be the bitstring representing a Slater determinant with Hamming weight $|\mathbf{x}| = N$. Let $\mathbf{e}_i \in \mathbb{F}_2^M$ be a bitstring with a ‘1’ in position i and ‘0’ everywhere else. $\mathbf{x}$ can be written as $\mathbf{x} = \mathbf{e}_{i_1} \oplus \mathbf{e}_{i_2} \oplus \dots \oplus \mathbf{e}_{i_N}$ with $i_1 < i_2 < \dots < i_N$. The sorted-list encoding $\mathcal{E}$ encodes $\mathbf{x}$ by concatenating together the binary representation of each of the occupied orbital indices in ascending order:

$$\mathcal{E}(\mathbf{x}) = |i_1\rangle \otimes |i_2\rangle \otimes \dots \otimes |i_N\rangle \otimes |\infty\rangle \otimes \dots \otimes |\infty\rangle.$$

With the total of M orbitals, each orbital index $|i\rangle$ can be represented as a binary string of length $\lceil \log_2 M \rceil$. The number of such registers N_{reg} does not necessarily needs be equal to the number of electrons N . A sentinel state representing “unoccupied electron” is used to fill in the empty registers when $N < N_{\text{reg}}$. By convention, the “unoccupied electron” state is represented by the symbol $|\infty\rangle$ and is ordered to be larger than all other orbital indices. Thus, $N_{\text{reg}} \lceil \log_2(M+1) \rceil$ qubits can hold all bitstrings of Hamming weight less than or equal to N_{reg} . This allows the sorted-list encoding to perform particle non-conserving operations as long as enough registers are set aside for the operation.

Before elaborating how fermionic operations are encoded, we first introduce several building block gates frequently used in the sorted-list encoding. The $= p$ gate (Figure A.1a) accepts one register as input and flips a target ancilla qubit if that register is equal to a constant value p . Similarly, the $< p$ gate (Figure A.2a) accepts one register as input and flips a target ancilla qubit if that register is equal to a constant value p . Both the $= p$ and $< p$ gates has indexed variants, shown in Figure A.1b and A.2b, respectively, for use in the qubitization circuits, where an additional register is added to accept a superposition of values for p . The $= p$ gate can be implemented using a multicontrolled Toffoli gate ($C^{\log M} X$ gate) while the implementation for the $< p$ gate can be found in Appendix H of a previous work [6]. The gate cost for both the constant and indexed variant of $= p$ and $< p$ are shown in Table A.1. We also have the bubble gate U_p (Figure A.3) which accepts two registers as input and swaps the two registers if one of them is equal to p and the other is larger than p . The original implementation of U_p uses $\log M$ controlled swap gates, and four $= p$ and $> p$ gates, where the $> p$ gate can be implemented similarly as the $< p$ gate [6]. For the purposes of our implementation, we introduced two implementations of the U_p gate specially designed for the trotterization and qubitization circuits, respectively, detailed in Section A.1 A.1.2 A.1.2.1. Finally, the $a \leftrightarrow b$ (Figure A.4) gate outputs b when the input is a and outputs a when the input is b , where both a and b are constant values. Otherwise the input remains unchanged.



Figure A.1: The $= p$ circuit [17]. This circuit flips ancilla $|a\rangle$ when the value of i is equal to (a) a constant p or (b) an indexed value p .



Figure A.2: The $< p$ circuit [17]. This circuit flips ancilla $|a\rangle$ when the value of i is less than (a) a constant p or (b) an indexed value p . While our work only uses the $< p$ gate, the original work includes the $\leq p$ and $> p$ variant which can be implemented similarly [6].

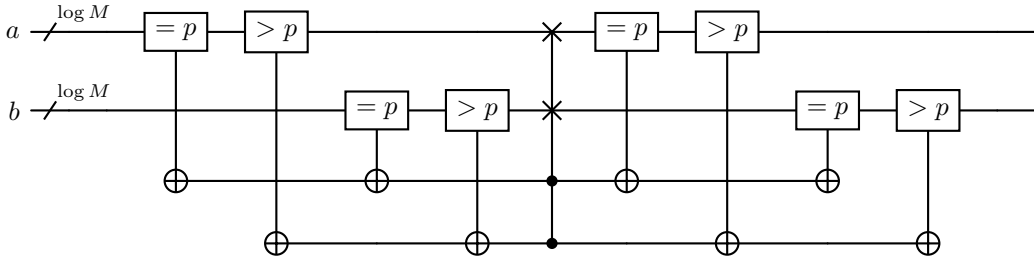


Figure A.3: The bubble circuit U_p [17]. This circuit swaps the two registers a and b when one of them is equal to p and the other is larger than p .

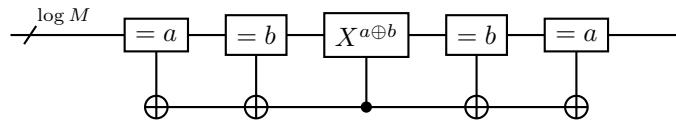


Figure A.4: The $a \leftrightarrow b$ circuit [17]. This circuit outputs b when the input is a and outputs a when the input is b . Otherwise, the input is unchanged. The $X^{a \oplus b}$ is composed of multiple CNOT gate according to the binary representation of $a \oplus b$.

Fermionic operations on this encoding are first decomposed to Majorana operators. The Majorana operators are then further decomposed to sgn-rank and bit-flip operators.

$$a_p^\dagger = \frac{1}{2}(\gamma_{p,0} - i\gamma_{p,1}), \quad a_p = \frac{1}{2}(\gamma_{p,0} + i\gamma_{p,1}),$$

$$\mathcal{E}(\gamma_{p,0}) = \text{bit-flip}(p) \text{sgn-rank}(p-1), \quad \mathcal{E}(\gamma_{p,1}) = i \text{bit-flip}(p) \text{sgn-rank}(p).$$

The $\text{sgn-rank}(p)$ operator calculates the parity of the number of occupied states for orbital indices less than or equal to p . It then multiplies the wavefunction by -1 for odd parity and $+1$ for even parity. On the other hand, the $\text{bit-flip}(p)$

operator flips the occupation of orbital p . The circuit implementations of the sgn-rank and bit-flip operators can be found in Figure A.5 and A.6, respectively. Essentially, the sgn-rank(j) circuit applies a Z gate for every register containing an orbital index less than or equal to j . On the other hand, bit-flip(j) first checks if j is occupied. If j is occupied it replaces $|j\rangle$ with $|\infty\rangle$, otherwise it replaces $|\infty\rangle$ with $|j\rangle$. After replacing the target register with the appropriate value, that register is sorted back to the correct ascending order. As the Majorana operators changes the number of occupied orbitals, the encoding needs to set aside more registers than the total number of electrons. More specifically, the one-electron terms will require 2 extra registers in the encoding than the total number of electrons, while the two-electron terms will require 4 extra registers. Such registers are filled with the $|\infty\rangle$ state so that the swap circuit in Figure A.6 can be done.

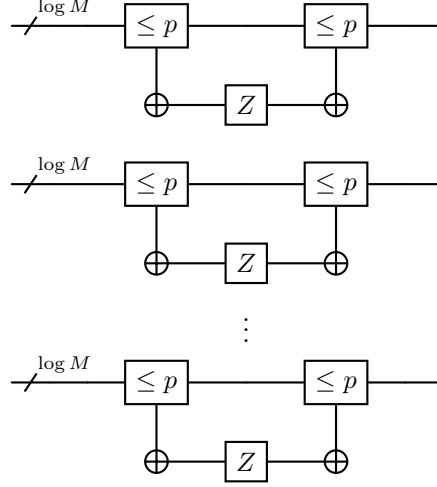


Figure A.5: The sgn-rank(p) circuit [17]. The $\leq p$ circuit is shown on Figure A.2a. Quantum wires not extending to both ends of the circuit are ancilla qubits.

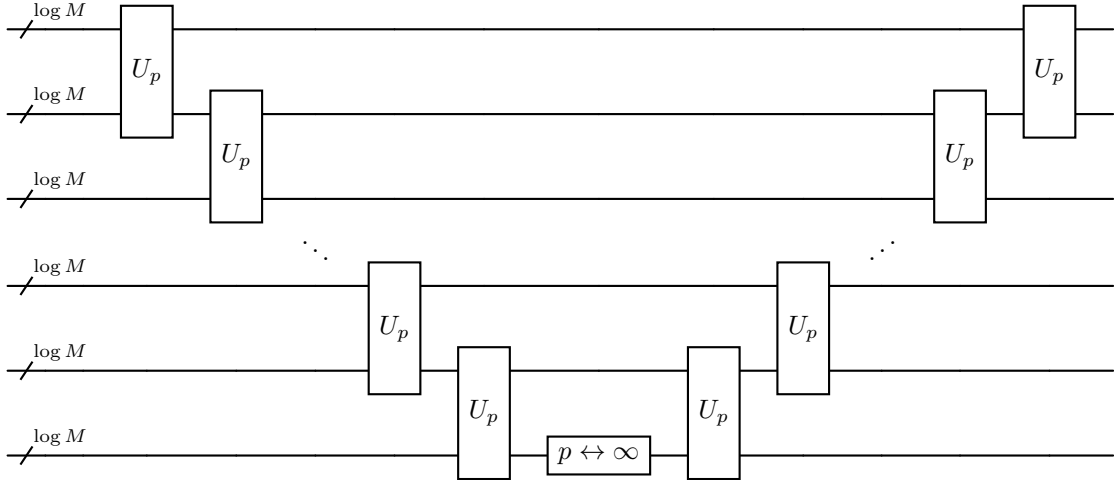


Figure A.6: The bit-flip(p) circuit [17]. The bubble circuit U_p is shown on Figure A.3, while the swap circuit $p \leftrightarrow \infty$ is shown in Figure A.4.

Table A.1: Costings of the constant and indexed variants of the $= p$ and $< p$ circuits using the implementation of [6]. $n_0(p)$ and $n_1(p)$ represents the number of ‘0’ and ‘1’ bits, respectively, in the binary representation of p .

Circuit	Constant	Indexed
$= p$	$C^{\log M} X$	1
Figure A.1	CX 0	$2 \log M$
	X $2n_0(p)$	$2 \log M$
$< p$	$C^2 X$ $\log M - 1$	$\log M$
Figure A.2	CX $6 \log M - 4$	$6 \log M - 5$
	X $2n_1(p)$	0

A.1.2.1. Additional Optimizations

Here, we show two optimizations we made on the bubble circuit (Figure A.3) denoted as U'_p and U''_p . The U'_p gate is shown in Figure A.7. In this case, we modified the circuit to use the $< p$ gate instead of the $> p$ gate. This only results in an open control for the swap gate as no two registers has the same value in the sorted-list encoding as long as $p \neq \infty$. This allows parity information of the two inputs to be optionally extracted into an ancilla qubit for parity calculations. Next, as the U_p circuit is applied sequentially to consecutive pairs of registers, the ancillae are reused so that some circuit elements can be omitted, as shown in Figure A.7 and A.8.

On the other hand, the U''_p gate precalculates $= p$ and $< p$ information of the registers in the beginning. This optimized bubble gate U''_p halves the number $= p$ and $< p$ gates, with the trade-off of increasing the CSWAP gate by 2 for every application of the U''_p gate, as shown in Figure A.9. As a result, the use of this circuit would add two additional ancillae to store $= p$ and $< p$ information for every register of the sorted-list encoding.

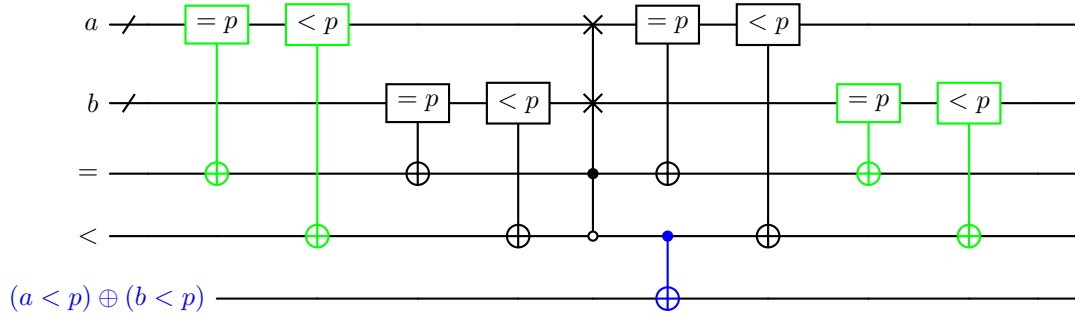


Figure A.7: The modified bubble circuit from Figure A.3. Here $< p$ is used instead of $> p$ so that parity information can also be optionally tapped as shown in blue. Finally, as this circuit is implemented sequentially to all consecutive pairs of registers, the $=$ and $<$ ancillae can be reused. As a result, the $= p$ and $< p$ at qubits a on the left and the $= p$ and $< p$ at qubits b on the right side shown in green can be omitted for the U'_p circuits in the middle as demonstrated in Figure A.8. This circuit has an input of size $2 \log M$ qubits and requires $2 \log M + 4$ ancillae.

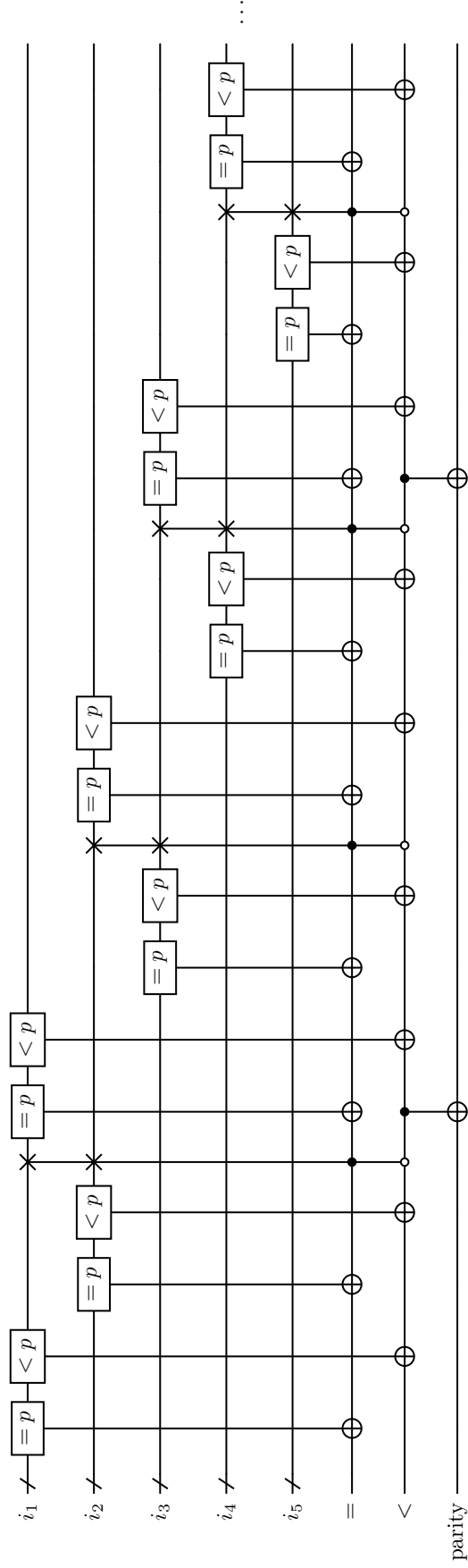


Figure A.8: This circuit shows the cascade of the modified bubble circuit U'_p using the same ancilla for the $=$ and $<$ calculation. As a result, some of the $= p$ and $< p$ gates can be omitted. The CNOT gate is also added to every other bubble circuit such that the parity qubit contains $(i_1 < p) \oplus (i_2 < p) \oplus \dots \oplus (i_N < p)$.

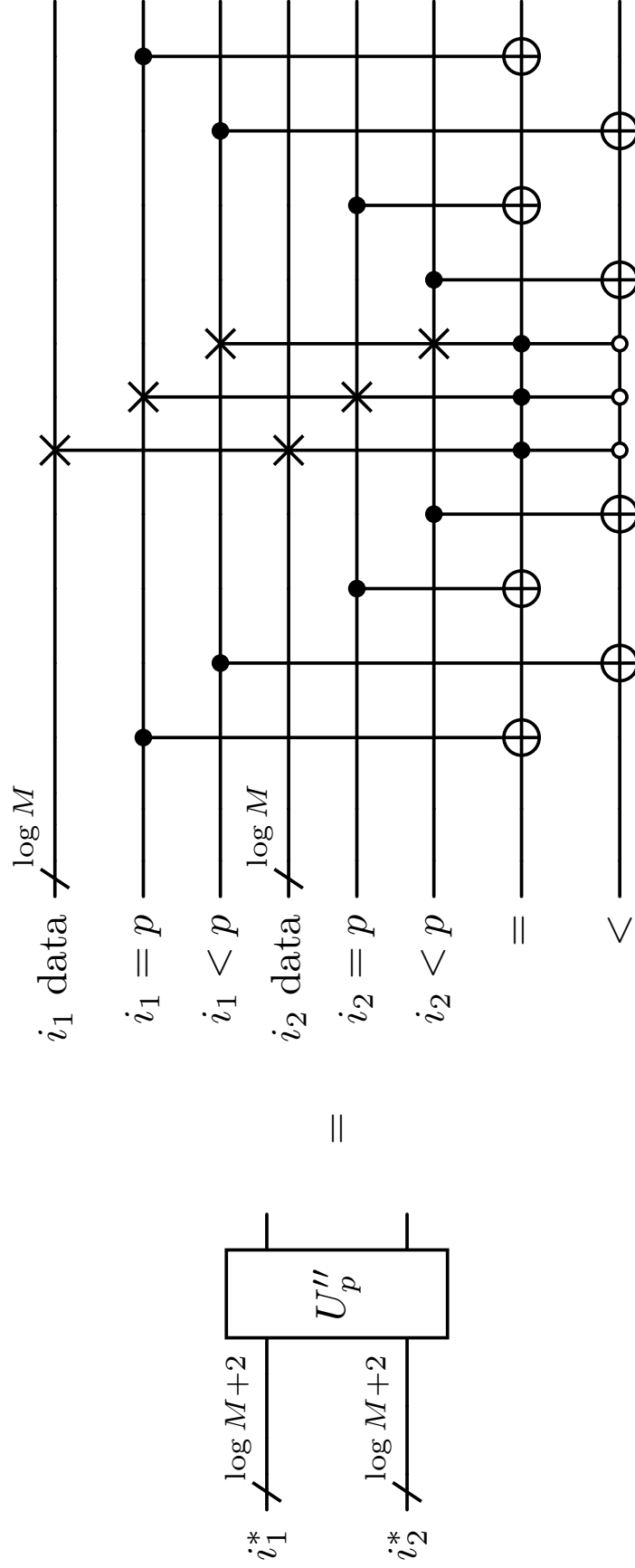


Figure A.9: The circuit implementing U_p'' , where the $=$ and $<$ gates are precalculated beforehand. The registers i_n^* contain $\log M$ qubits storing the binary representation of the orbital indices (shown as i_n data), along with 2 additional ancilla qubits to store the precalculated $=$ and $<$ data (shown as $i_n = p$ and $i_n < p$, respectively).

Appendix A.2: Hamiltonian Simulation

Numerous quantum algorithms have been proposed to calculate the electronic structure of quantum chemical systems. This includes Variational Quantum Eigensolver (VQE) [40–43], quantum imaginary time evolution algorithms [44, 45], quantum Monte Carlo methods [46], Quantum Phase Estimation (QPE) [47, 48], among others. In our paper, we focus on the implementation of the Quantum Phase Estimation (QPE) algorithm, which is one of the fundamental quantum algorithms to calculate the eigenvalues of a unitary circuit. As typical quantum chemical Hamiltonians are not unitary, it cannot be directly used with the QPE algorithm. Instead, a proxy operator, implementable in quantum circuits, is constructed such that the Hamiltonian’s eigenvalues can be obtained from the eigenvalues of the proxy operator.

A.2.1. Robust Phase Estimation with Trotterization

One example of the proxy operator is the Hamiltonian evolution e^{-iHt} , which is often implemented via trotterization [24, 49]. The eigenvalues of the evolution, $E_k(e^{-iHt})$, are related to the eigenvalues of the Hamiltonian, $E_k(H)$, by the following equation:

$$E_k(e^{-iHt}) = \exp(-iE_k(H)t).$$

Robust Phase Estimation (RPE) [2, 50–53] is particularly well-suited for use with Trotter formulas due to the low number of ancilla qubits required. The RPE algorithm performs phase estimation with only one ancilla qubit, typically via Hadamard tests, as depicted in Figure A.10. Given an error parameter ε_{qpe} , we perform the Hadamard test to obtain the expectation value of e^{-iHt} at different values of t :

$$\begin{aligned} g(t) &= \langle \Psi | e^{-iHt} | \Psi \rangle; \quad t = 2^1, \dots, 2^{n_{\text{qpe}}} \\ &= \sum_k |\langle \Psi | \Psi_k \rangle|^2 e^{-iE_k t}. \end{aligned}$$

By interpreting $g(t)$ as a time signal, the ground-state energy can be extracted as the lowest frequency component via signal processing techniques [2]. To obtain the ground-state energy with precision ε , we require $\mathcal{O}(1/\varepsilon\delta)$ applications of the trotter formula $S(\delta)$ that approximates the Hamiltonian evolution $e^{-iH\delta}$. In this case, the trotter time step δ is chosen to ensure the trotter error is bounded by $\varepsilon\delta$.

Implementing $S(\delta)$ involves decomposing the Hamiltonian into a linear combination of Hermitian operators H_l , such that each individual $e^{-iw_l H_l \delta_l}$ can be implemented as a quantum circuit:

$$H = \sum_{l=0}^{L-1} w_l H_l. \quad (\text{A.1})$$

For standard encodings such as the Jordan-Wigner, Parity, and Bravyi-Kitaev [11, 26], the individual Hermitian terms are typically chosen to be Pauli strings.

The time evolution $e^{-iH\delta}$ can then be approximated as a product of individual $e^{-iw_l H_l \delta_l}$ terms, with multiple variations of the exponential product formulas available offering varying levels of accuracy. One such formula is the deterministic p^{th} -order Suzuki formula [24] shown below:

$$\begin{aligned} e^{-iH\delta} &\approx S_p^{(\text{det})}(\delta), \\ S_1^{(\text{det})}(\delta) &= \prod_{l=0}^{L-1} e^{-iw_l H_l \delta}, \\ S_2^{(\text{det})}(\delta) &= \prod_{l=0}^{L-1} e^{-iw_l H_l \delta/2} \prod_{l=L-1}^{0*} e^{-iw_l H_l \delta/2}, \\ S_{p+2}^{(\text{det})}(\delta) &= (S_p^{(\text{det})}(s_p \delta))^2 S_p^{(\text{det})}((1 - 4s_p)\delta) (S_p^{(\text{det})}(s_p \delta))^2, \end{aligned} \quad (\text{A.2})$$

where the product $\prod^*$ is taken in reverse order and $s_p = (4 - \sqrt[p+1]{4})^{-1}$. Another approach is the random qDRIFT formula [25]:

$$e^{-iH\delta} \approx S_N^{(\text{rand})}(\delta) = \prod_{l=1}^N e^{-i\lambda H_l \delta/N}.$$

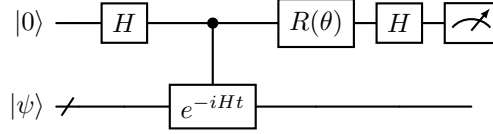


Figure A.10: The Hadamard test circuit to obtain $\langle \psi | e^{-iHt} | \psi \rangle$. The angle θ of the phase gate $R(\theta)$ is set to 0 for $\text{Re} \langle \psi | e^{-iHt} | \psi \rangle$ and $\pi/2$ for $\text{Im} \langle \psi | e^{-iHt} | \psi \rangle$.

In this case, the index j is chosen randomly using a probability distribution $p_l = h_l/\lambda$ and N is chosen based on the precision parameter ε . Finally, the partially random formula partitions the Hamiltonian into deterministic and random implementations:

$$e^{-iHt} \approx e^{-iH_{\text{deter}}t} e^{-iH_{\text{rand}}t}.$$

Table A.2 presents the total cost of performing the RPE calculation to obtain the ground-state of a Hamiltonian with precision ε using these three Trotter formulas.

Table A.2: The number of applications of the time evolution of the elementary Hermitians operators $e^{-i w_l H_l t}$ required to obtain the ground-state with precision ε using the various Trotter product formulas. These costs are obtained from a previous work [2]. Here, L is the number of Hermitian terms, λ is the 1-norm of the Hamiltonian.

Formula	Cost
Deterministic p^{th} -order [24]	$\mathcal{O}(5^{p/2-1} L^{2+1/p} / \varepsilon^{1+1/p})$
Random qDRIFT [25]	$\mathcal{O}(\lambda^2 / \varepsilon^2)$
Partially random [2]	$\mathcal{O}(5^{p/2-1} L^{2+1/p} / \varepsilon^{1+1/p} + \lambda^2 / \varepsilon^2)$

A.2.1.1. Trotterization for the Plane-Wave Basis

As mentioned in the main text, a lower gate cost can be achieved with the plane-wave basis due to the inherent structure of the plane-wave Hamiltonian. We define ν as the indices for the plane-wave (momentum) basis, and p as indices for the plane-wave dual (position) basis. These two bases are related by the Fourier transform:

$$|\nu\rangle = \frac{1}{\sqrt{M}} \sum_p |p\rangle e^{-ik_\nu \cdot r_p}.$$

The same relationship holds in the second quantization for creation and annihilation operators, linking the plane-wave basis operators ($a_\nu^\dagger/a_p$) and the plane-wave dual basis operators ($c_p^\dagger/c_p$):

$$a_\nu^\dagger = \frac{1}{\sqrt{M}} \sum_p c_p^\dagger e^{-ik_\nu \cdot r_p}.$$

Conversion between the two bases can be performed using the quantum Fourier transform (QFT) circuit [54, 55] for the first quantization, and the fermionic fast Fourier transform (FFFT) [18] for the second quantization.

The Hamiltonian can then be written as:

$$H = \text{FT}^\dagger \hat{T} \text{FT} + \hat{U} + \hat{V}, \quad (\text{A.3})$$

where $\hat{T}$, $\hat{U}$, and $\hat{V}$ represents the kinetic, external, and coulomb operators, respectively, expressed in first and second

quantization as follows:

$$\begin{aligned}
\hat{T}^{(1)} &= \sum_i^N \sum_\nu^M T_\nu |\nu\rangle\langle\nu|_i, & \hat{T}^{(2)} &= \sum_\nu^M T_\nu a_\nu^\dagger a_\nu, \\
\hat{U}^{(1)} &= \sum_i^N \sum_p^M U_p |p\rangle\langle p|_i, & \hat{U}^{(2)} &= \sum_p^M U_p n_p, \\
\hat{V}^{(1)} &= \sum_{i \neq j}^N \sum_{p \neq q}^M V_{pq} |p\rangle\langle p|_i |q\rangle\langle q|_j, & \hat{V}^{(2)} &= \sum_{p \neq q}^M V_{pq} n_p n_q,
\end{aligned}$$

where $n_p = c_p^\dagger c_p$ denotes the number operator in the dual basis. Furthermore, the operator FT represents applying QFT for all the electronic degrees of freedom for the first quantization, or the FFFT operator [18] for the second quantization.

The terms $|\nu\rangle\langle\nu|_i$ are diagonal in the plane-wave basis, while $|p\rangle\langle p|_i$ and $|p\rangle\langle p|_i |q\rangle\langle q|_j$ are diagonal in the plane-wave dual basis. The same property holds for their second-quantized counterparts, allowing all the kinetic operators to mutually commute in the plane-wave basis and all the potential operators to mutually commute in the plane-wave dual basis. This mutually commuting property is highly advantageous for trotterization, as the time evolution of $\hat{T}$ and $\hat{U} + \hat{V}$ can be implemented exactly, without any intrinsic trotter errors.

$$\begin{aligned}
e^{-i\hat{T}^{(1)}\delta} &= \prod_i^N e^{-i\hat{T}_i^{(1)}\delta} = \prod_i^N e^{-i\delta \sum_\nu^M T_\nu |\nu\rangle\langle\nu|_i}, \\
e^{-i(\hat{U}^{(1)} + \hat{V}^{(1)})\delta} &= \prod_i^N e^{-i\hat{U}_i^{(1)}\delta} \prod_{i \neq j}^N e^{-i\hat{V}_{ij}^{(1)}\delta} = \prod_i^N e^{-i\delta \sum_p^M U_p |p\rangle\langle p|_i} \prod_{i \neq j}^N e^{-i\delta \sum_{p \neq q}^M V_{pq} |p\rangle\langle p|_i |q\rangle\langle q|_j}, \\
e^{-i\hat{T}^{(2)}\delta} &= \prod_\nu^M e^{-i\delta T_\nu a_\nu^\dagger a_\nu}, \\
e^{-i(\hat{U}^{(2)} + \hat{V}^{(2)})\delta} &= \prod_p^M e^{-i\delta U_p n_p} \prod_{p \neq q}^M e^{-i\delta V_{pq} n_p n_q}.
\end{aligned}$$

Thus, we effectively have $L = 2$ for the Hamiltonian decomposition, comprising $\hat{T}$ and $\hat{U} + \hat{V}$. Due to the low value of L , it is evident that the deterministic formula is more efficient for this Hamiltonian. As an example, the second-order ($p = 2$) Suzuki-Trotter product formula can be written as:

$$S_2(\delta) = e^{-i(\hat{U} + \hat{V})\delta/2} \text{FT}^\dagger e^{-i\hat{T}\delta} \text{FT} e^{-i(\hat{U} + \hat{V})\delta/2}.$$

A.2.2. Szegedy Quantum Walk with Qubitization

A more recent method of constructing the proxy operator for phase estimation algorithms is through qubitization. In this approach, the Szegedy walk operator [18], implementable as a quantum circuit, serves as the proxy unitary. The construction of this walk operator can be summarized as follows. Unlike the trotterization case, the Hamiltonian terms are divided into a linear combination of unitaries, where all the coefficients w_l are positive (with any sign or complex phase incorporated into the unitary itself):

$$H = \sum_{l=0}^{L-1} w_l U_l \quad w_l \in \mathbb{R}^+.$$

Those terms are then incorporated into the Szegedy walk operator using the PREPARE and SELECT circuits, defined in the high-level as:

$$\text{PREPARE } |0\rangle^{\otimes \lceil \log L \rceil} = \sum_{l=0}^{L-1} \sqrt{\frac{|w_l|}{\lambda}} |l\rangle^{\otimes \lceil \log L \rceil}, \quad (\text{A.4})$$

$$\text{SELECT } |l\rangle \otimes |\psi\rangle = |l\rangle \otimes U_l |\psi\rangle. \quad (\text{A.5})$$

The PREPARE circuit prepares the coefficient magnitudes $|w_l|$ on $\lceil \log_2 L \rceil$ ancilla qubits, where $|l\rangle^{\otimes \lceil \log L \rceil}$ is the binary representation of the index l , and $\lambda = \sum_{l=0}^{L-1} |w_l|$. On the other hand, the SELECT circuit applies the unitaries U_l based on the indices l encoded on the ancilla qubits. At this point, applying a composition of these two operators yields a normalized representation of the Hamiltonian:

$$\text{PREPARE}^\dagger \cdot \text{SELECT} \cdot \text{PREPARE} |0\rangle \otimes |\psi\rangle = |0\rangle \otimes \frac{H}{\lambda} |\psi\rangle. \quad (\text{A.6})$$

However, Equation A.6 cannot be directly used for the QPE algorithm because H is not unitary, as previously noted. The Szegedy walk operator [18] is constructed instead:

$$\mathcal{W} = (2 \text{PREPARE} |0\rangle\langle 0| \text{PREPARE}^\dagger \otimes \mathbb{I} - \mathbb{I} \otimes \mathbb{I}) \cdot \text{SELECT}. \quad (\text{A.7})$$

The eigenvalues of the walk operator, $E_k(\mathcal{W})$, and the Hamiltonian, $E_k(H)$, are then related by:

$$E_k(\mathcal{W}) = \pm \arccos(E_k(H)/\lambda). \quad (\text{A.8})$$

Consequently, one can obtain $E_k(H)$ by performing the QPE algorithm on $\mathcal{W}$. For this, we opt to use the optimal QPE algorithm, which minimizes the number of $\mathcal{W}$ applications [18, 56], shown in Figure 2 of [18].

Various implementations exist for the PREPARE and SELECT circuits. Implementations for second-quantized Hamiltonians in the MO basis, using the 8-fold symmetry of the coulomb terms, include methods like single-factorization [6], double-factorization [7], and tensor hypercontraction [8]. These techniques lead to circuit implementations with varying T gate complexities, where lower scaling often comes with the trade-off of more expensive classical preprocessing. Alternatively, implementations that exploit the inherent structure of the plane-wave basis set typically utilizes the first-quantized encoding [15, 16].

The following section highlighted the single-factorization method [6], used for the simulation of the second-quantized Hamiltonian in the MO basis-set, along with the first-quantized plane-wave algorithm [15, 16]. This serves as an introduction to both methods as they will be incorporated into our circuit formulations of the sorted-list encoding.

A.2.2.1. Single-Factorization

The single-factorization algorithm [6] focuses on reducing the complexity of implementing the two-electron terms in a quantum circuit. Although it is possible to construct spin-unrestricted implementations of this algorithm, only the spin-restricted version is discussed in previous works [6–8]. Furthermore, all the h_{pq} and h_{pqrs} coefficients must be real and that the h_{pqrs} term must have 8-fold symmetry:

$$\begin{aligned} h_{pq} &\in \mathbb{R}, \quad h_{pqrs} \in \mathbb{R}, \\ h_{pq} &= h_{qp}, \\ h_{pqrs} &= h_{prqs} = h_{sqrp} = h_{srqp}, \\ &= h_{qpsr} = h_{qspr} = h_{rpsq} = h_{rspq}. \end{aligned} \quad (\text{A.9})$$

As a result, the more compact molecular orbital basis sets are typically used instead of the plane-wave basis set. Despite this, other basis sets such as the Daubechies wavelets [57] can also be used as long as the constraints of Equations A.9 are satisfied.

Here, we show a summary of the single-factorization algorithm [6] along with the optimizations made in another work [8]. While they directly converted the fermionic operators to Pauli operators using the Jordan-Wigner encoding, we opt to convert the fermionic operators to Majorana operators instead. First, an alternative second-quantized representation of the Hamiltonian, shown in Equation A.10, is used:

$$H_{\text{SF}} = \sum_{\sigma \in \{\uparrow, \downarrow\}} \sum_{p,q=1}^{M/2} T_{pq} a_{p,\sigma}^\dagger a_{q,\sigma} + \frac{1}{2} \sum_{\alpha, \beta \in \{\uparrow, \downarrow\}} \sum_{p,q,r,s=1}^{M/2} V_{pqrs} a_{p,\alpha}^\dagger a_{q,\alpha} a_{r,\beta}^\dagger a_{s,\beta}. \quad (\text{A.10})$$

Next, we define the operator $Q_{pq\sigma}$ in terms of Majorana operators:

$$\begin{aligned} Q_{pq\sigma} &= i\gamma_{p,0,\sigma}\gamma_{q,1,\sigma}, \\ a_{p,\sigma}^\dagger a_{q,\sigma} + a_{q,\sigma}^\dagger a_{p,\sigma} &= i\frac{\gamma_{p,0,\sigma}\gamma_{q,1,\sigma}}{2} + i\frac{\gamma_{q,0,\sigma}\gamma_{p,1,\sigma}}{2}, \\ a_{p,\sigma}^\dagger a_{p,\sigma} &= \frac{\mathbb{I}}{2} + i\frac{\gamma_{p,0,\sigma}\gamma_{p,1,\sigma}}{2}. \end{aligned}$$

Due to the symmetries of Equation A.9, we can then express H_{SF} in terms of $Q_{pq\sigma}$. Note that the constant $\mathbb{I}/2$ term in the decomposition of $a_{p,\sigma}^\dagger a_{p,\sigma}$ is not implemented by the $Q_{pq\sigma}$ operator and therefore must be explicitly accounted for, resulting in T'_{pq} and a constant term:

$$\begin{aligned} H_{\text{SF}} &= H_{\text{const}} \mathbb{I} + \frac{1}{2} \sum_{\sigma \in \{\uparrow, \downarrow\}} \sum_{p,q=1}^{M/2} T'_{pq} Q_{pq\sigma} + \frac{1}{8} \sum_{\alpha, \beta \in \{\uparrow, \downarrow\}} \sum_{p,q,r,s=1}^{M/2} V_{pqrs} Q_{pq\alpha} Q_{rs\beta}, \\ H_{\text{const}} &= \sum_{p=1}^{M/2} T_{pp} + \frac{1}{2} \sum_{pr=1}^{M/2} V_{pprr}, \\ T'_{pq} &= T_{pq} + \sum_{r=1}^{M/2} V_{pqrr}. \end{aligned}$$

Next, the $M/2 \times M/2 \times M/2 \times M/2$ tensor V_{pqrs} is converted to an $M^2/4 \times M^2/4$ matrix W , with the rows containing the composite index pq and the columns containing the composite index rs ($W_{pM/2+q, rM/2+s} = V_{pqrs}$). This matrix is then diagonalized to obtain eigenvalues w_l and eigenvectors $g_{pq}^{(l)}$. As a result,

$$\frac{1}{8} \sum_{\alpha, \beta} \sum_{p,q,r,s=1}^{M/2} V_{pqrs} Q_{pq\alpha} Q_{rs\beta} = \sum_{l=1}^L w_l \left(\sum_{\sigma \in \{\uparrow, \downarrow\}} \sum_{p,q=1}^{M/2} g_{pq}^{(l)} Q_{pq\sigma} \right)^2.$$

In this case, only the largest $L \ll M^2/4$ eigenvalues of W is included, chosen such that this differs from the original Hamiltonian by an acceptable error, with empirical observations showing that $L \sim \mathcal{O}(M \log M)$ [58]. After the T'_{pq} , w_l , $g_{pq}^{(l)}$ coefficients are calculated, the PREPARE circuit is implemented as follows:

$$\begin{aligned} \text{PREPARE} &\rightarrow |0\rangle_a \sum_{pq\sigma} \sqrt{\frac{|T'_{pq}|}{2\lambda_{\text{SF}}}} \left| \tau_{pq}^{(0)} \right\rangle_b |0\rangle_c |p; q; \sigma\rangle_d |0\rangle_e \\ &+ \sum_l \sqrt{\frac{w_l}{\lambda_{\text{SF}}}} |l\rangle_a \sum_{pqrs\alpha\beta} \sqrt{|g_{pq}^{(l)} g_{rs}^{(l)}|} \left| \tau_{pq}^{(l)} \right\rangle_b \left| \tau_{rs}^{(l)} \right\rangle_c |p; q; \alpha\rangle_d |r; s; \beta\rangle_e. \end{aligned} \quad (\text{A.11})$$

The preparation of the coefficients is done using the coefficient oracle developed in [18], with the QROM implemented using the techniques that trade dirty or clean qubits to reduce its T gate cost [27]. On the other hand, the SELECT circuit is implemented as two circuits below. SELECT₁ is applied for any value of $|l\rangle_a$, while SELECT₂ is conditionally applied when $|l\rangle_a$ is nonzero:

$$\text{SELECT}_1 |\tau\rangle_b |p; q; \alpha\rangle_d |\psi\rangle = |\tau\rangle_b |p; q; \alpha\rangle_d \otimes (-1)^\tau i \gamma_{p,0,\alpha} \gamma_{q,1,\alpha} |\psi\rangle, \quad (\text{A.12})$$

$$\text{SELECT}_2 |\tau\rangle_c |r; s; \beta\rangle_e |\psi\rangle = |\tau\rangle_c |r; s; \beta\rangle_e \otimes (-1)^\tau i \gamma_{r,0,\beta} \gamma_{s,1,\beta} |\psi\rangle. \quad (\text{A.13})$$

One final issue to account for is the conversion from the spatial-orbital indices to the spin-orbital indices. This is done using a simple controlled addition circuit [59]. Table A.3 details the indices a to e of the PREPARE circuit.

Table A.3: Legend for the qubit label of Equation A.11.

Qubit Label	Description
a	Indices of the eigenvalues and eigenvectors of the matrix W . $ 0\rangle_a$ is reserved for the one-electron term.
b and c	Sign of the T'_{pq} for the one-electron term, and the sign of the $g_{pq}^{(l)}$ and $g_{rs}^{(l)}$, respectively, for the two-electron term.
d and e	Orbital indices and spin for the Majorana operators.

A.2.2.2. First-Quantized Encoding

To leverage the good structure of the plane-wave basis, we divide the Hamiltonian into the kinetic $\hat{T}$, external potential $\hat{U}$, and coulomb potential $\hat{V}$ terms. In the first-quantization, those can be written as:

$$H^{(1)} = \hat{T}^{(1)} + \hat{U}^{(1)} + \hat{V}^{(1)}, \quad (\text{A.14})$$

$$\hat{T}^{(1)} = \sum_{i=1}^N \left(\sum_{p \in G} \frac{\|k_p\|^2}{2} |p\rangle\langle p|_i \right), \quad (\text{A.15})$$

$$\hat{U}^{(1)} = -\frac{4\pi}{\Omega} \sum_{i=1}^N \sum_{\substack{\nu \in G \\ \nu \neq 0}} \left(\sum_{\substack{p \in G \\ (p-\nu) \in G}} \sum_{I=1}^{N_{at}} Z_I \frac{e^{-ik_\nu \cdot \mathbf{X}_I}}{\|k_\nu\|^2} |p-\nu\rangle\langle p|_i \right), \quad (\text{A.16})$$

$$\hat{V}^{(1)} = \frac{2\pi}{\Omega} \sum_{\substack{i,j=1 \\ i \neq j}}^N \sum_{\substack{\nu \in G \\ \nu \neq 0}} \left(\sum_{\substack{p,q \in G \\ (p+\nu) \in G \\ (q-\nu) \in G}} \frac{1}{\|k_\nu\|^2} |p+\nu\rangle\langle p|_i |q-\nu\rangle\langle q|_j \right). \quad (\text{A.17})$$

The $\hat{T}^{(1)}$, $\hat{U}^{(1)}$, $\hat{V}^{(1)}$ terms can be decomposed into a linear combination of unitary terms (after normalization), shown as brackets in the equation above. The action of such unitaries (with the normalization omitted for the sake of clarity) on an arbitrary Hartree product can be described by the following equations.

$$\begin{aligned} |a\rangle_1 \cdots |p\rangle_i \cdots |z\rangle_N &\xrightarrow{T \text{ unitary}} \frac{\|k_p\|^2}{2} (|a\rangle_1 \cdots |p\rangle_i \cdots |z\rangle_N), \\ |a\rangle_1 \cdots |p\rangle_i \cdots |z\rangle_N &\xrightarrow{U \text{ unitary}} \left(\sum_{I=1}^{N_{at}} Z_I \frac{e^{-ik_\nu \cdot \mathbf{X}_I}}{\|k_\nu\|^2} \right) (|a\rangle_1 \cdots |p-\nu\rangle_i \cdots |z\rangle_N), \\ |a\rangle_1 \cdots |p\rangle_i \cdots |q\rangle_j \cdots |z\rangle_N &\xrightarrow{V \text{ unitary}} \frac{1}{\|k_\nu\|^2} (|a\rangle_1 \cdots |p+\nu\rangle_i \cdots |q-\nu\rangle_j \cdots |z\rangle_N). \end{aligned}$$

Taking advantage of such unitary formulation, the PREPARE circuit has the form:

$$\begin{aligned} \text{PREPARE} &\rightarrow (\cos \theta |0\rangle + \sin \theta |1\rangle)_a |+\rangle_b \frac{1}{\sqrt{N}} \left(\sqrt{N-1} |0\rangle_c \sum_{i \neq j=1}^N |i\rangle_d |j\rangle_e + |1\rangle_c \sum_{j=1}^N |j\rangle_d |j\rangle_e \right) \\ &\left(\frac{1}{\sqrt{3}} \sum_{w=0}^2 |w\rangle_f \right) \left(\frac{1}{2^{n_p-1}-1} \sum_{r,s=0}^{n_p-2} 2^{(r+s)/2} |r\rangle_g |s\rangle_h \right) \left(\sqrt{\frac{\lambda_U}{\lambda_U + \lambda_V}} |0\rangle_i + \sqrt{\frac{\lambda_V}{\lambda_U + \lambda_V}} |1\rangle_i \right) \\ &\left(\sqrt{\frac{p_\nu}{\lambda_\nu}} |0\rangle_j \sum_{\nu \in G_0} \frac{1}{\|\nu\|} |\nu\rangle_k + \sqrt{1-p_\nu} |1\rangle_j |\nu^\perp\rangle_k \right) \left(\frac{1}{\sqrt{\sum_I Z_I}} \sum_{I=1}^{N_{at}} \sqrt{Z_I} |I\rangle_l \right), \quad (\text{A.18}) \end{aligned}$$

where the qubit labels a to l are explained in Table A.4. Moreover, the SELECT circuit has the following form below:

$$\begin{aligned} \text{SELECT}_{\hat{T}^{(1)}} |b\rangle_b |j\rangle_e |w\rangle_f |r\rangle_g |s\rangle_h |p_j\rangle &= (-1)^{b(p_w, r p_w, s \oplus 1)} |b\rangle_b |j\rangle_e |w\rangle_f |r\rangle_g |s\rangle_h |p_j\rangle, \\ \text{SELECT}_{\hat{U}^{(1)}} |b\rangle_b |j\rangle_e |0\rangle_i |\nu\rangle_k |I\rangle_l |q_j\rangle &= -e^{-ik_\nu \cdot \mathbf{X}_I} (-1)^{b[(p-\nu) \notin G]} |b\rangle_b |j\rangle_e |0\rangle_i |\nu\rangle_k |I\rangle_l |q_j - \nu\rangle, \\ \text{SELECT}_{\hat{V}^{(1)}} |b\rangle_b |i\rangle_d |j\rangle_e |1\rangle_i |\nu\rangle_k |p_i\rangle |q_j\rangle &= (-1)^{b([(p+\nu) \notin G] \vee [(q-\nu) \notin G])} |b\rangle_b |i\rangle_d |j\rangle_e |1\rangle_i |\nu\rangle_k |p_i + \nu\rangle |q_j - \nu\rangle. \end{aligned} \quad (\text{A.19})$$

Here, we separate the circuit based on the application of $\hat{T}^{(1)}$, $\hat{U}^{(1)}$, and $\hat{V}^{(1)}$. Furthermore, the indices inside the ket such as $|p_i\rangle$ and $|q_j\rangle$ refers to the wavefunction at the i^{th} and j^{th} registers (electronic degrees of freedom), respectively. On the other hand, the indices outside the ket corresponds ancillas generated by the PREPARE circuit of Equation A.18 and explained in Table A.4.

Table A.4: Legend for the qubit label of Equation A.18.

Qubit Label	Description
a	$ 0\rangle_a$ tells the SELECT circuit to apply $\hat{T}^{(1)}$ $ 1\rangle_a$ tells the SELECT circuit to apply $\hat{U}^{(1)} + \hat{V}^{(1)}$
b	Used to cancel out the unitary application when $(p + \nu) \notin G$ and when $(q - \nu) \notin G$ for $\hat{U}^{(1)}$ and $\hat{V}^{(1)}$ (see Equation A.16 and A.17)
c	Specifies if $i = j$ for the case of $\hat{V}^{(1)}$ (see Equation A.17)
d and e	The electron indices i and j , respectively
w	$ 0\rangle_w, 1\rangle_w, 2\rangle_w$ specifies the Cartesian axes x, y , and z , respectively, used for the SELECT circuit to apply $\hat{T}^{(1)}$
g and h	Used for the application of $\ k_p\ ^2$ in $\hat{T}^{(1)}$ (see Equation A.15)
i	$ 1\rangle_a 0\rangle_i$ tells the SELECT circuit to apply $\hat{U}^{(1)}$ $ 1\rangle_a 1\rangle_i$ tells the SELECT circuit to apply $\hat{V}^{(1)}$
j and k	Creates the $1/\ k_\nu\ ^2$ coefficients for $\hat{U}^{(1)}$ and $\hat{V}^{(1)}$ (see Equation A.16 and A.17)
l	The ion indices I for $\hat{U}^{(1)}$ (see Equation A.16)

Appendix A.3: Trotterization Implementation for the Sorted-List Encoding

A.3.1. Circuit Implementations

For our trotterization implementation of the sorted-list encoding, the Hamiltonian evolution is decomposed into the following unitaries:

$$\exp(i\theta(a_p^\dagger a_q + a_q^\dagger a_p)), \quad (\text{A.20})$$

$$\exp(i\theta(a_p^\dagger a_q - i a_q^\dagger a_p)), \quad (\text{A.21})$$

$$\exp(i\theta(a_p^\dagger a_q a_r a_s + a_s^\dagger a_r^\dagger a_q a_p)), \quad (\text{A.22})$$

$$\exp(i\theta(a_p^\dagger a_q a_r a_s - i a_s^\dagger a_r^\dagger a_q a_p)). \quad (\text{A.23})$$

Notice that $(a_p^\dagger a_q \pm a_q^\dagger a_p)^2$ can be written as:

$$\begin{aligned}
(a_p^\dagger a_q \pm a_q^\dagger a_p)^2 &= (a_p^\dagger a_q)^2 + (a_q^\dagger a_p)^2 \pm a_p^\dagger a_q a_q^\dagger a_p \pm a_q^\dagger a_p a_p^\dagger a_q \\
&= \pm(a_p^\dagger a_q a_q^\dagger a_p + a_q^\dagger a_p a_p^\dagger a_q), \\
(a_p^\dagger a_q + a_q^\dagger a_p)^2 |\psi\rangle &= \begin{cases} |\psi\rangle & \text{if } a_p^\dagger a_q |\psi\rangle \neq 0 \text{ or } a_q^\dagger a_p |\psi\rangle \neq 0 \\ 0 & \text{otherwise} \end{cases}, \\
(i(a_p^\dagger a_q - a_q^\dagger a_p))^2 |\psi\rangle &= \begin{cases} |\psi\rangle & \text{if } a_p^\dagger a_q |\psi\rangle \neq 0 \text{ or } a_q^\dagger a_p |\psi\rangle \neq 0 \\ 0 & \text{otherwise} \end{cases}.
\end{aligned}$$

Thus, given any $|\psi\rangle$ such that either $a_p^\dagger a_q |\psi\rangle \neq 0$ or $a_q^\dagger a_p |\psi\rangle \neq 0$ (either orbital p is occupied, or orbital q is occupied, but not both), we can expand the exponent of Equation A.20 into the Taylor series:

$$\begin{aligned}
\exp(i\theta(a_p^\dagger a_q + a_q^\dagger a_p)) &= \left[\left(\sum_{n=0}^{\infty} \frac{(i\theta)^{2n}}{2n!} \right) + \left(\sum_{n=0}^{\infty} \frac{(i\theta)^{2n+1}}{(2n+1)!} \right) (a_p^\dagger a_q + a_q^\dagger a_p) \right] |\psi\rangle \\
&= (\cos \theta + i \sin \theta (a_p^\dagger a_q + a_q^\dagger a_p)) |\psi\rangle.
\end{aligned}$$

If $|\psi\rangle$ does not satisfy the previous condition, it reverts to the identity operator. Through similar arguments, we can obtain similar identities for Equation A.21 to A.23.

$$\begin{aligned}
\exp(i\theta(a_p^\dagger a_q + a_q^\dagger a_p)) |\psi\rangle &= \begin{cases} (\cos \theta + i \sin \theta(a_p^\dagger a_q)) |\psi\rangle & \text{if } a_p^\dagger a_q |\psi\rangle \neq 0 \\ (\cos \theta + i \sin \theta(a_q^\dagger a_p)) |\psi\rangle & \text{if } a_q^\dagger a_p |\psi\rangle \neq 0 \\ |\psi\rangle & \text{otherwise} \end{cases} \\
\exp(i\theta(ia_p^\dagger a_q - ia_q^\dagger a_p)) |\psi\rangle &= \begin{cases} (\cos \theta + i \sin \theta(ia_p^\dagger a_q)) |\psi\rangle & \text{if } a_p^\dagger a_q |\psi\rangle \neq 0 \\ (\cos \theta + i \sin \theta(-ia_q^\dagger a_p)) |\psi\rangle & \text{if } a_q^\dagger a_p |\psi\rangle \neq 0 \\ |\psi\rangle & \text{otherwise} \end{cases} \\
\exp(i\theta(a_p^\dagger a_q^\dagger a_r a_s + a_s^\dagger a_r^\dagger a_q a_p)) |\psi\rangle &= \begin{cases} (\cos \theta + i \sin \theta(a_p^\dagger a_q^\dagger a_r a_s)) |\psi\rangle & \text{if } a_p^\dagger a_q^\dagger a_r a_s |\psi\rangle \neq 0 \\ (\cos \theta + i \sin \theta(a_s^\dagger a_r^\dagger a_q a_p)) |\psi\rangle & \text{if } a_s^\dagger a_r^\dagger a_q a_p |\psi\rangle \neq 0 \\ |\psi\rangle & \text{otherwise} \end{cases} \\
\exp(i\theta(ia_p^\dagger a_q^\dagger a_r a_s - ia_s^\dagger a_r^\dagger a_q a_p)) |\psi\rangle &= \begin{cases} (\cos \theta + i \sin \theta(ia_p^\dagger a_q^\dagger a_r a_s)) |\psi\rangle & \text{if } a_p^\dagger a_q^\dagger a_r a_s |\psi\rangle \neq 0 \\ (\cos \theta + i \sin \theta(-ia_s^\dagger a_r^\dagger a_q a_p)) |\psi\rangle & \text{if } a_s^\dagger a_r^\dagger a_q a_p |\psi\rangle \neq 0 \\ |\psi\rangle & \text{otherwise} \end{cases}
\end{aligned}$$

The circuit implementing both Equation A.20 and A.21 is shown in Figure A.11. Looking at the right-hand side of Equation A.20 and A.21, the circuit would apply a Pauli evolution if p or q is occupied, but not both. Otherwise, it will perform the identity operation. This is achieved by first using the U'_p and U'_q gates of Figure A.7 to bubble down both the p and q orbitals to the last register. Parity calculations are also carried out by tapping the ancilla qubits of some of the U'_p and U'_q gates. At this point, register i_N would contain either p or q , if orbital p or q is occupied, respectively. The $=p$ and $=q$ gates are then used to check whether orbital p or q are occupied. If only one of orbital p or q is occupied, we would apply an evolution gate swapping between the binary representation of p and q on register i_N , as shown in Equation A.24. This controlled evolution gate also includes another ancilla that calculates the parity, which we will elaborate further towards the end of this section. Figure A.16 shows an example of the evolution gate applied on register i_N and the parity ancilla, written as:

$$\text{ROT}(\theta) = \exp(i\theta(X_{i_N}^{p \oplus q} \otimes Z_{\text{parity}})). \quad (\text{A.24})$$

The circuit implementing Equation A.21 is done almost identically, with one of the X gates in Equation A.24 replaced with the Y gate.

The circuit implementing the two electron unitaries of Equation A.22 and A.23 is shown in Figure A.12. This is done similarly with the one-electron case, just that the p and s indices are bubbled down to register i_N , while the q and r indices are bubbled down to register i_{N-1} . Parity calculations are again carried out by tapping the ancilla qubits of some of the bubble gates. The final adjustments to parity are made using the P gate which is a CNOT gate controlled by one of the $=p$, $=q$, $=r$, or $=s$ ancillae, depending on the exact value of the indices. Finally, we have the following controlled rotation gate applied on the second last register, the last register, and the parity ancilla, written as:

$$\text{ROT}(\theta) = \exp(i\theta(X_{i_{N-1}}^{q \oplus r} \otimes X_{i_N}^{p \oplus s} \otimes Z_{\text{parity}})). \quad (\text{A.25})$$

This Pauli rotation gate is applied when both p and q are occupied and both r and s are unoccupied or when both r and s are occupied and both p and q are unoccupied as shown with the Toffoli gates targeting the ancilla qubit labelled “control”.

The circuits of Figure A.11 and A.12 can be slightly simplified when there are common indices, mainly:

$$\exp(i\theta a_p^\dagger a_p), \exp(i\theta a_p^\dagger a_q^\dagger a_q a_p), \text{ and } \exp(i\theta(ia_p^\dagger a_q^\dagger a_q a_s - ia_s^\dagger a_q^\dagger a_q a_p)).$$

Hence, we show specialized circuits for the following in Figure A.13-A.15. The structure of the circuits of Figure A.11 and A.12 results in one more opportunity for optimization. Suppose that we have two successive unitary operations with common indices as follows:

$$e^{i\theta_2(a_p^\dagger a_r + a_r^\dagger a_p)} e^{i\theta_1(a_p^\dagger a_q + a_q^\dagger a_p)}.$$

Due to the common orbital index p , the U'_p gate on the right side of the $e^{i\theta_1(a_p^\dagger a_q + a_q^\dagger a_p)}$ circuit cancels out with the U'_p gate on the left side of the $e^{i\theta_2(a_p^\dagger a_r + a_r^\dagger a_p)}$. As a result, the two U'_p gates can be omitted from the circuit.

One last thing to consider is the parity calculation. The parity calculation can be explained by first looking into the Jordan-Wigner encoding of the fermionic operators [21]

$$a_p^\dagger a_q = X_p X_q \left(\frac{I + Z_p}{2} \right) \left(\frac{I - Z_q}{2} \right) \Theta_{pq} \left(\prod_{i=0}^{p-1} Z_i \right) \left(\prod_{i=0}^{q-1} Z_i \right), \quad (\text{A.26})$$

$$\begin{aligned} a_p^\dagger a_q^\dagger a_r a_s &= X_p X_q X_r X_s \left(\frac{I + Z_p}{2} \right) \left(\frac{I + Z_q}{2} \right) \left(\frac{I - Z_r}{2} \right) \left(\frac{I - Z_s}{2} \right) \\ &\times \Theta_{pq} \Theta_{pr} \Theta_{ps} \Theta_{qr} \Theta_{qs} \Theta_{rs} \left(\prod_{i=0}^{p-1} Z_i \right) \left(\prod_{i=0}^{q-1} Z_i \right) \left(\prod_{i=0}^{r-1} Z_i \right) \left(\prod_{i=0}^{s-1} Z_i \right), \end{aligned} \quad (\text{A.27})$$

where we have $\Theta_{ij} = +1$ if $i \leq j$ and $\Theta_{ij} = -1$ if $i > j$. The equation above can be divided into the parity, update, and projection terms [21]. The X gates are the update terms, the $(I \pm Z)/2$ gates are the projection terms, and finally the Θ_{ij} and the $\prod_{i=0}^p Z_i$ are the parity terms. Since the $\prod_{i=0}^p Z_i$ terms are the same between $a_p^\dagger a_q$ and $a_q^\dagger a_p$ and between $a_p^\dagger a_q^\dagger a_r a_s$ and $a_s^\dagger a_r^\dagger a_q a_p$, those terms can be implemented as follows:

$$\left(\prod_{i=0}^{p-1} Z_i \right) \left(\prod_{i=0}^{q-1} Z_i \right) \Rightarrow (i_1 < p) \oplus \dots \oplus (i_N < p) \oplus (i_1 < q) \oplus \dots \oplus (i_N < q), \quad (\text{A.28})$$

$$\begin{aligned} \left(\prod_{i=0}^{p-1} Z_i \right) \left(\prod_{i=0}^{q-1} Z_i \right) \left(\prod_{i=0}^{r-1} Z_i \right) \left(\prod_{i=0}^{s-1} Z_i \right) &\Rightarrow (i_1 < p) \oplus \dots \oplus (i_N < p) \oplus (i_1 < q) \oplus \dots \oplus (i_N < q) \\ &\oplus (i_1 < r) \oplus \dots \oplus (i_N < r) \oplus (i_1 < s) \oplus \dots \oplus (i_N < s). \end{aligned} \quad (\text{A.29})$$

Such information can be tapped from the ancilla of the bubble gate U' as shown in Figure A.7. On the other hand, we also have the Θ terms that differ between $a_p^\dagger a_q$ and $a_q^\dagger a_p$ and between $a_p^\dagger a_q^\dagger a_r a_s$ and $a_s^\dagger a_r^\dagger a_q a_p$, respectively. Those are

$$\Theta_{pq} \quad \text{and} \quad \Theta_{pq} \Theta_{pr} \Theta_{ps} \Theta_{qr} \Theta_{qs} \Theta_{rs}, \quad (\text{A.30})$$

respectively. As a result, those terms are implemented as a CNOT gate with one of the $= p, = q, = r$, or $= s$ ancillae as the control and the parity ancilla as the target as shown in Figure A.11 and the P gate in Figure A.12.

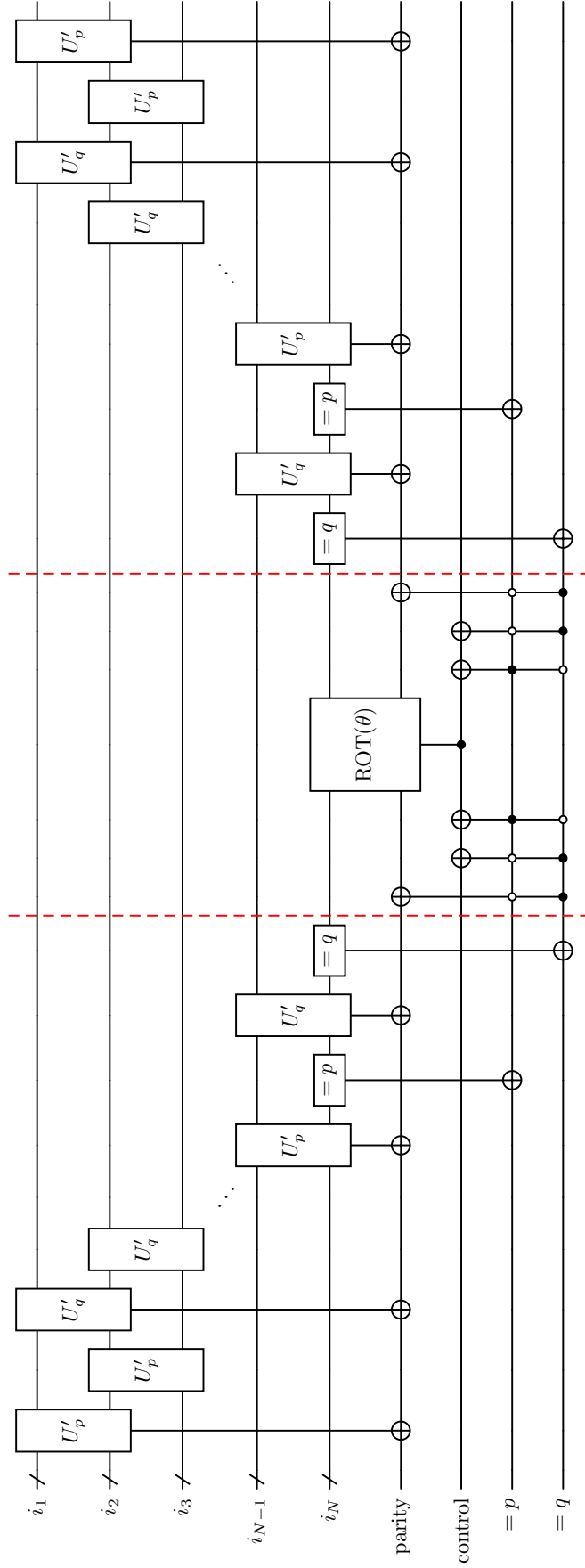


Figure A.11: The circuit implementing $\exp(i\theta(X_{i_N}^{p \oplus q} \otimes Z_{parity}))$ and $\exp(i\theta(ia_p^\dagger a_q - ia_q^\dagger a_p))$ for $p > q$. For $\exp(i\theta(a_p^\dagger a_q + a_q^\dagger a_p))$, $\text{ROT}(\theta)$ is a controlled Pauli evolution gate $\exp(i\theta(X_{i_N}^{p \oplus q} \otimes Z_{parity}))$, where X gates are placed in the '1' bits of the binary representation of $p \oplus q$, and a Z gate is placed on the parity ancilla. The $\text{ROT}(\theta)$ for $\exp(i\theta(ia_p^\dagger a_q - ia_q^\dagger a_p))$ is implemented similarly, with one of the X replaced by a Y . Both can be implemented efficiently using the techniques in [60]. Note that the $= p$ gate is implicitly done in the U'_p cascade. This is achieved by using the $=$ ancilla in U'_p of Figure A.7 as the $= p$ ancilla here and omitting the $= p$ gate in the final U'_p gate of i_{N-1} and i_N . The $= q$ gate is also done implicitly with the U'_q cascade. The parity calculation is performed by obtaining parity information from the U'_p and U'_q gates as shown in Figure A.7 such that the parity ancilla contains the expression of Equation A.28 just after the p and q indices are bubbled down. The red dashed line separates the circuit into the BUBBLE operation, the ROTATE operation, and the BUBBLE † operation from left to right. In total, this circuit acts on $N \log M$ data qubits and requires $N(\log M + 1) + 6$ ancillae.

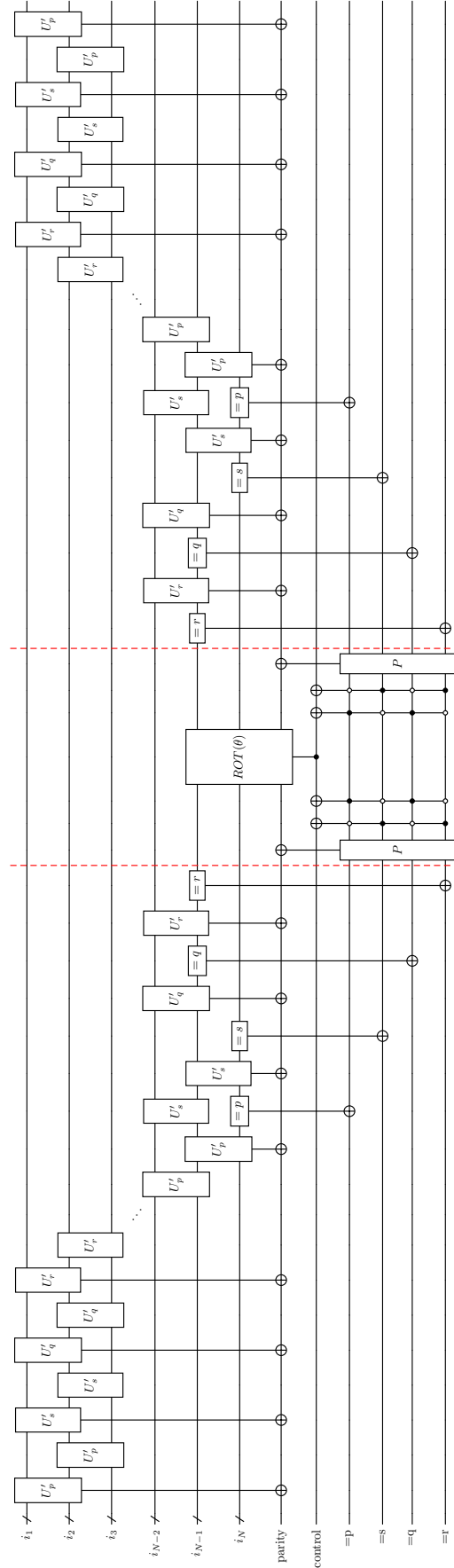


Figure A.12: The circuit implementing $\exp(i\theta(a_p^\dagger a_q^\dagger a_r a_s + a_s^\dagger a_r^\dagger a_q a_p))$ and $\exp(i\theta(a_p^\dagger a_q^\dagger a_r a_s - ia_s^\dagger a_r^\dagger a_q a_p))$. The gate P flips the parity qubit according to Equation A.30. For $\exp(i\theta(a_p^\dagger a_q^\dagger a_r a_s + a_s^\dagger a_r^\dagger a_q a_p))$, $\text{ROT}(\theta)$ is a controlled Pauli evolution gate $\exp(i\theta X_{i_{N-1}}^{p \oplus s} X_{i_N}^{p \oplus s} \otimes Z_{\text{parity}})$, where X gates are placed in the '1' bits of the binary representation of $q \oplus r$ and $p \oplus s$, and a Z gate is placed on the parity ancilla. The $\text{ROT}(\theta)$ for $\exp(i\theta(ia_p^\dagger a_q^\dagger a_r a_s - ia_s^\dagger a_r^\dagger a_q a_p))$ is implemented similarly, with one of the X replaced by a Y . Both can be implemented efficiently using the techniques in [60]. The parity calculation is done by obtaining parity information from the U'_p, U'_q, U'_r, U'_s gates as shown in Figure A.7 such that the parity ancilla contains the expression of Equation A.29 just after the p, q, r , and s indices are bubbled down. The red dashed line separates the circuit into the BUBBLE operation, the ROTATE operation, and the BUBBLE † operation from left to right. In total, this circuit acts on $N \log M$ data qubits and requires $N(\log M + 1) + 10$ ancillae.

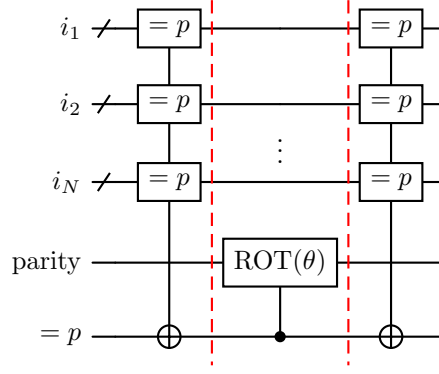


Figure A.13: This circuit implements $\exp(i\theta a_p^\dagger a_p)$. The red dashed line separates the circuit into the BUBBLE operation, the ROTATE operation, and the BUBBLE † operation from left to right. In total, this circuit acts on $N \log M$ qubits and requires $N(\log M - 1) + 2$ ancillae.

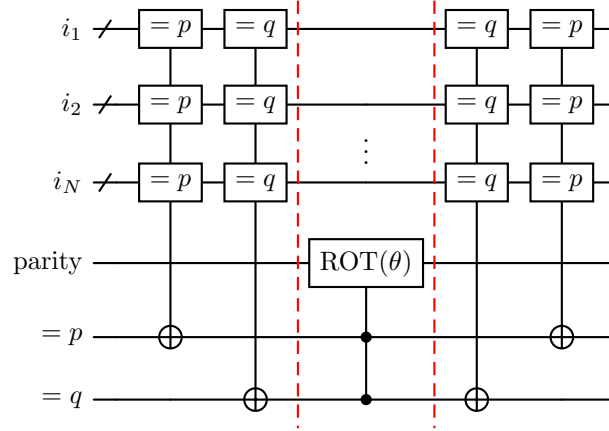


Figure A.14: This circuit implements $\exp(i\theta a_p^\dagger a_q^\dagger a_q a_p)$. The red dashed line separates the circuit into the BUBBLE operation, the ROTATE operation, and the BUBBLE † operation from left to right. In total, this circuit acts on $N \log M$ qubits and requires $N(\log M - 1) + 3$ ancillae.

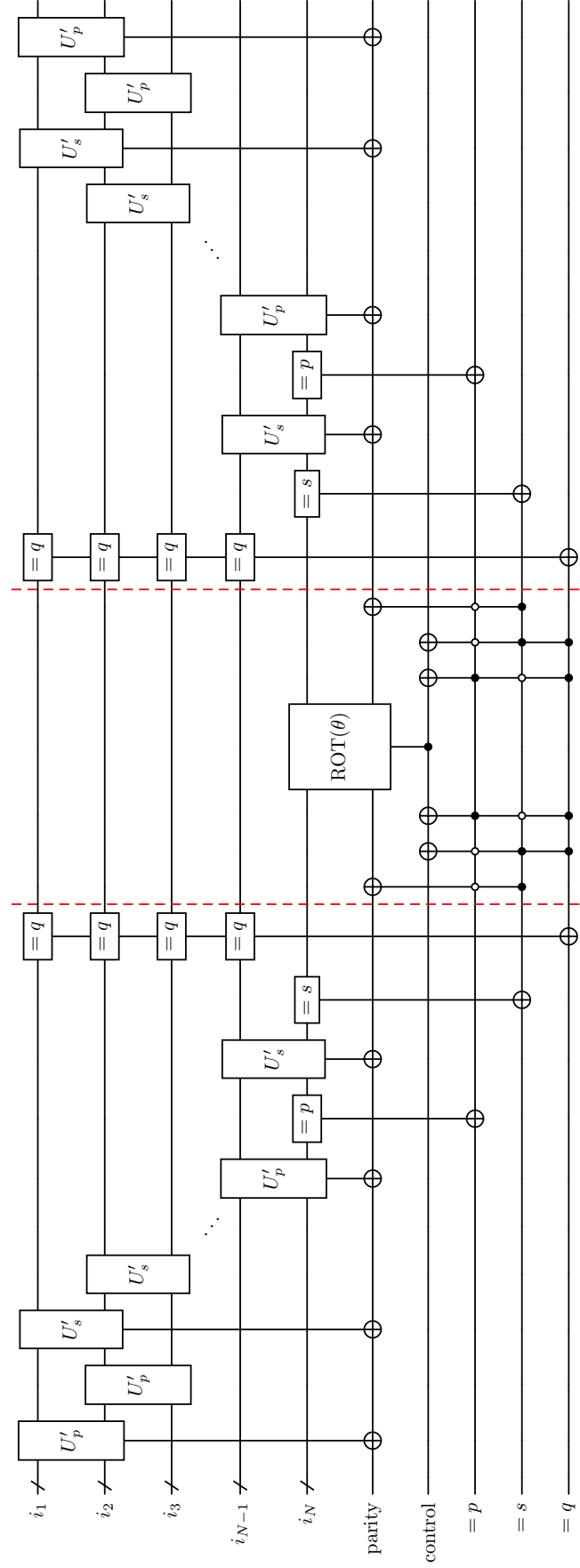


Figure A.15: This circuit implements $\exp(i\theta(a_p^\dagger a_q a_s + a_s^\dagger a_q a_p))$ and $\exp(i\theta(a_p^\dagger a_q a_s - ia_s^\dagger a_q a_p))$. The red dashed line separates the circuit into the BUBBLE operation, the ROTATE operation, and the BUBBLE † operation from left to right. In total, this circuit acts on $N \log M$ qubits and requires $N(\log M + 1) + 7$ ancillae.

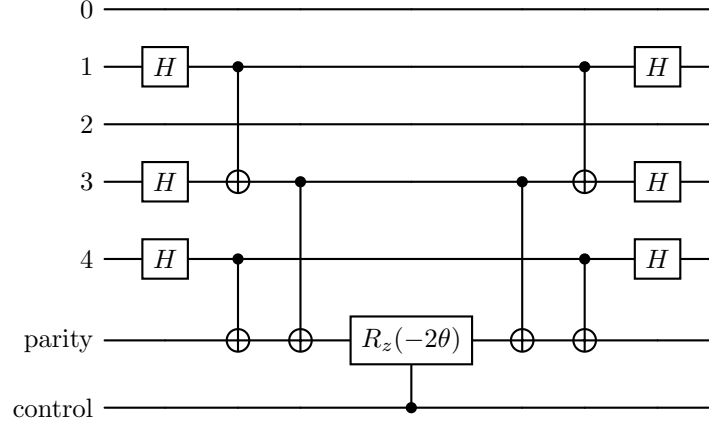


Figure A.16: The controlled rotation circuit implementing $\exp(i\theta X_{p\oplus q} \otimes Z_{\text{parity}})$, where $p \oplus q$ has binary representation of 11010. We used the CNOT cascade techniques discussed in [60].

A.3.2. Cost Analysis

Table A.5: Costings of the building blocks of the trotterization circuits. $n_0(p)$ and $n_1(p)$ represents the number of ‘0’ and ‘1’ bits, respectively, in the binary representation of p . The circuit Bubble_p^N shown in (Figure A.17f) is composed of $(2N - 1) [= p]$ gates, $2N [< p]$ gates, $(2N - 2)$ X gates, $(N - 1) \log M$ CSWAP gates, and $\frac{1}{2}N + \frac{1}{2}(N \bmod 2)$ CX gates.

Circuit	Costings
Rot_p^1 (Fig. A.17a)	CRZ 1
Rot_{pq}^2 (Fig. A.17b)	C^2X 1 CRZ 1
Rot_{pq}^1 (Fig. A.17c)	C^2X 4 CX $2n_1(p \oplus q) + 2$ X 8 H $2n_1(p \oplus q)$ CRZ 1
Rot_{pqs}^2 (Fig. A.17d)	C^3X 4 CX $2n_1(p \oplus s) + 2$ X 8 H $2n_1(p \oplus s)$ CRZ 1
Rot_{pqrs}^2 (Fig. A.17e)	C^4X 4 CX $2[n_1(p \oplus s) + n_1(q \oplus r)] + 4$ X 16 H $2[n_1(p \oplus s) + n_1(q \oplus r)]$ CRZ 1
Bubble_p^N (Fig. A.17f)	$C^{\log M}X$ $2N - 1$ CSWAP $N \log M - N$ C^2X $2N \log M - 2N$ CX $12N \log M - 7.5N + 0.5(N \bmod 2)$ X $(4N - 2)n_0(p) + 4Nn_1(p) + 2N - 2$

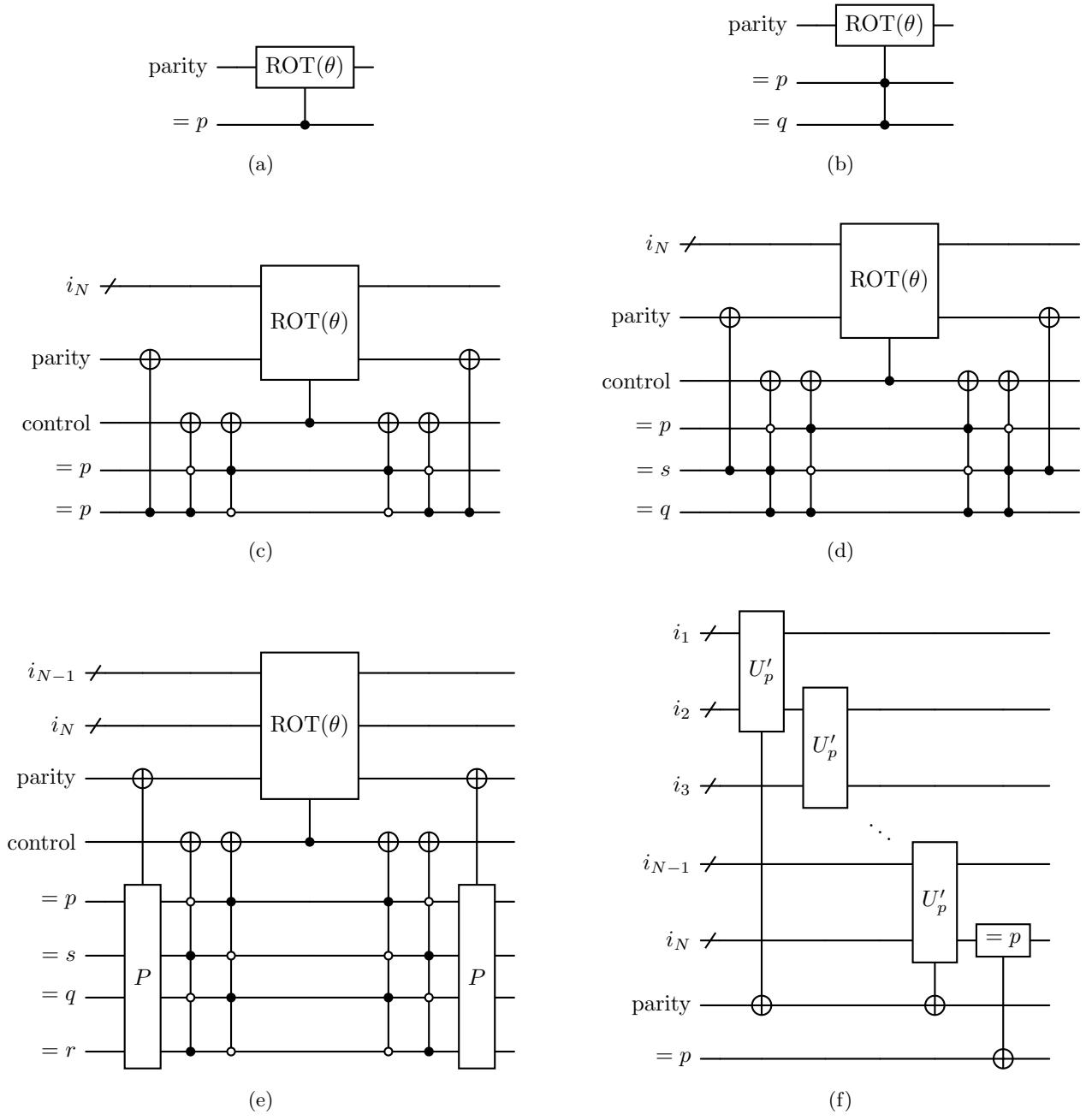


Figure A.17: Building blocks of the Trotterization circuits with costings shown in Table A.5

Table A.6: Costings of the trotterization circuits. $n_0(p)$ and $n_1(p)$ represents the number of ‘0’ and ‘1’ bits, respectively, in the binary representation of p .

Circuit	Costings
$e^{i\theta(a_p^\dagger a_q + a_q^\dagger a_p)}$ and $e^{i\theta(i a_p^\dagger a_q - i a_q^\dagger a_p)}$ (Figure A.11)	Blocks $2[\text{Bubble}_p^N] + 2[\text{Bubble}_q^N] + [\text{Rot}_{pq}^1]$ $C^{\log M} X$ $8N - 4$ CSWAP $4N \log M - 4N$ $C^2 X$ $8N \log M - 8N + 4$ CX $48N \log M - 30N + 2(N \bmod 2) + 2 + 2n_1(p \oplus q)$ X $(8N - 4)[n_0(p) + n_0(q)] + 8N[n_1(p) + n_1(q)] + 8N$ H $2n_1(p \oplus q)$ CRZ 1
$e^{i\theta a_p^\dagger a_p}$ (Figure A.13)	Blocks $2N[=p] + [\text{Rot}_p^1]$ $C^{\log M} X$ $2N$ CRZ 1 X $2Nn_0(p)$
$e^{i\theta(a_p^\dagger a_q^\dagger a_r a_s + a_s^\dagger a_r^\dagger a_q a_p)}$ and $e^{i\theta(i a_p^\dagger a_q^\dagger a_r a_s - i a_s^\dagger a_r^\dagger a_q a_p)}$ (Figure A.12)	Blocks $2[\text{Bubble}_p^N] + 2[\text{Bubble}_s^N] + 2[\text{Bubble}_q^{N-1}] + 2[\text{Bubble}_r^{N-1}] + [\text{Rot}_{pqrs}^2]$ $C^{\log M} X$ $16N - 6$ $C^4 X$ 4 CSWAP $8N \log M - 8N - 4 \log M + 4$ $C^2 X$ $16N \log M - 16N - 8 \log M + 8$ CX $96N \log M - 60N - 48 \log M + 36 + 2[n_1(p \oplus s) + n_1(q \oplus r)]$ X $16N - 8 + (8N - 4)[n_0(p) + n_0(s)] + 8N[n_1(p) + n_1(s)]$ $\quad + (8N - 12)[n_0(q) + n_0(r)] + (8N - 8)[n_1(q) + n_1(r)]$ H $2[n_1(p \oplus s) + n_1(q \oplus r)]$ CRZ 1
$e^{i\theta(a_p^\dagger a_q^\dagger a_q a_s + a_s^\dagger a_q^\dagger a_q a_p)}$ and $e^{i\theta(i a_p^\dagger a_q^\dagger a_q a_s - i a_s^\dagger a_q^\dagger a_q a_p)}$ (Figure A.15)	Blocks $2[\text{Bubble}_p^N] + 2[\text{Bubble}_s^N] + 2(N - 1)[=q] + [\text{Rot}_{pq}^2]$ $C^{\log M} X$ $10N - 6$ $C^3 X$ 4 CSWAP $4N \log M - 4N$ $C^2 X$ $8N \log M - 8N$ CX $48N \log M - 30N - 2(N \bmod 2) + 2 + 2n_1(p \oplus s)$ X $(8N - 4)[n_0(p) + n_0(s)] + 8N[n_1(p) + n_1(s)] + 8N$ $\quad + (4N - 4)n_0(q)$ H $2n_1(p \oplus s)$ CRZ 1
$e^{i\theta a_p^\dagger a_q^\dagger a_q a_p}$ (Figure A.14)	Blocks $2N[=p] + 2N[=q] + [\text{Rot}_{pq}^2]$ $C^{\log M} X$ $4N$ $C^2 X$ 2 X $4N[n_0(p) + n_0(q)]$ CRZ 1

A.3.3. Simulation Methods for Trotterization cost

As mentioned in the main manuscript, tight theoretical error bounds are challenging to obtain. As a result, we used experimental error bounds obtained from a previous work [2] that uses the RPE algorithm. We used Equation E4 in the appendix of that work [2] for the deterministic trotter formulas, reproduced here below with the variables explained in Table A.7:

$$G = 5\pi N_{\text{stage}} L_{\text{det}} G_{\text{avg}} \sqrt{\frac{(p+1)^{1+1/p}}{p}} \frac{C_{\text{gs}}^{1/p}}{\varepsilon^{1+1/p}}. \quad (\text{A.31})$$

Of particular note is the constant C_{gs} which is obtained from empirical calculations. For our calculations, we used Figure 8 of [2] to obtain the following formula with respect to the 1-norm of the Hamiltonian λ :

$$C_{\text{gs}} = 3.470 \times 10^{-5} \lambda^{2.081}.$$

On the other hand, the cost for the randomized trotter formula is as follows:

$$G = G_{\text{avg}} \frac{16.3\lambda^2}{\varepsilon^2}, \quad (\text{A.32})$$

with the variables explained in Table A.7. Finally, the partially randomized formula has the following costs:

$$G = 30G_{\text{avg}}N_{\text{stage}}L_{\text{det}}e^{2/\kappa} \frac{0.1\pi}{\delta\sqrt{\varepsilon^2 - C_{\text{gs}}^2\delta^4}} + \frac{280}{9}G_{\text{avg}}\kappa e^{2/\kappa} \frac{(0.1\pi\lambda_{\text{rand}})^2}{\varepsilon^2 - C_{\text{gs}}^2\delta^4}, \quad (\text{A.33})$$

where the variable κ is minimized analytically using a quadratic formula, and the variable δ is minimized numerically.

Table A.7: Legend for the parameters of Equations A.31-A.33.

Parameters	Description
ε	Target precision for the energy, set to chemical accuracy (1.6×10^{-3} Ha)
G	Total gate costs
L_{det}	Number of deterministic trotter terms
δ	Trotter step size
p	Order of the trotter formula, set to 2 (Suzuki-Trotter) for our calculations
N_{stage}	$2 \times 5^{p/2-1}$, set to 2 (Suzuki-Trotter) for our calculations
C_{gs}	Trotter error constant with respect to the ground state energy
G_{avg}	Average gate costs to implement one unitary
κ	Scaling factor for the number of evolution terms for the randomized trotter formula
λ_{rand}	1-norm of terms treated randomly for the partially random trotter formula

Appendix A.4: Qubitization Implementation for the Sorted-List Encoding

A.4.1. Circuit Implementations

Here, we show the circuit implementation and cost analysis of the SELECT circuit. While a Majorana-free SELECT circuit can be constructed with similar gate cost scaling, we instead opt to implement Majorana operators for our qubitization circuit due to its simpler construction and lower T gate cost. The Majorana-free version of this SELECT circuit can be found in Section A.4 A.4.5. As we are applying Majorana operators for our circuit, the encoding needs to have at least $N+4$ registers to accomodate the two-electron terms. Additionally, we make use of the U_p'' optimizations of the bubble gate detailed in Section A.1 A.1.2 A.1.2.1. This requires 2 additional ancilla qubits for every register to hold temporary values during each Majorana operations.

For reference, we use the following form of the Majorana operators:

$$\gamma_{p,0} = a_p^\dagger + a_p, \quad \gamma_{p,1} = i(a_p^\dagger - a_p).$$

The SELECT circuit then applies four Majorana operators based on the indices of the ancilla qubits:

$$\begin{aligned} & \text{SELECT } |\theta_p; p; \theta_q; q; 0; 0; 0; 0\rangle |\psi\rangle \\ &= |\theta_p; p; \theta_q; q; 0; 0; 0; 0\rangle \gamma_{p,\theta_p} \gamma_{q,\theta_q} |\psi\rangle, \\ & \text{SELECT } |\theta_p; p; \theta_q; q; \theta_r; r; \theta_s; s\rangle |\psi\rangle \\ &= |\theta_p; p; \theta_q; q; \theta_r; r; \theta_s; s\rangle \gamma_{p,\theta_p} \gamma_{q,\theta_q} \gamma_{r,\theta_r} \gamma_{s,\theta_s} |\psi\rangle. \end{aligned}$$

To distinguish between the one- and two-electron terms, we can either use controlled variants of the $\gamma_{r,\theta_r} \gamma_{s,\theta_s}$ terms or by setting $\theta_r = \theta_s = 0$ and $r = s$ when applying the one-electron terms. Figure A.18 shows the circuit that implements SELECT, with the circuit of the individual Majorana operator γ_{p,θ_p} shown in Figure A.19. Like in the trotterization case, the $\mathcal{O}(N \log M)$ scaling is maintained with respect to the qubit count (including the ancillae) and gate count (including non-Toffoli gates).

Figure A.18: Circuit implementing SELECT, made up of four blocks of the γ_{p,θ_p} circuits, with the individual blocks defined in Figure A.19.

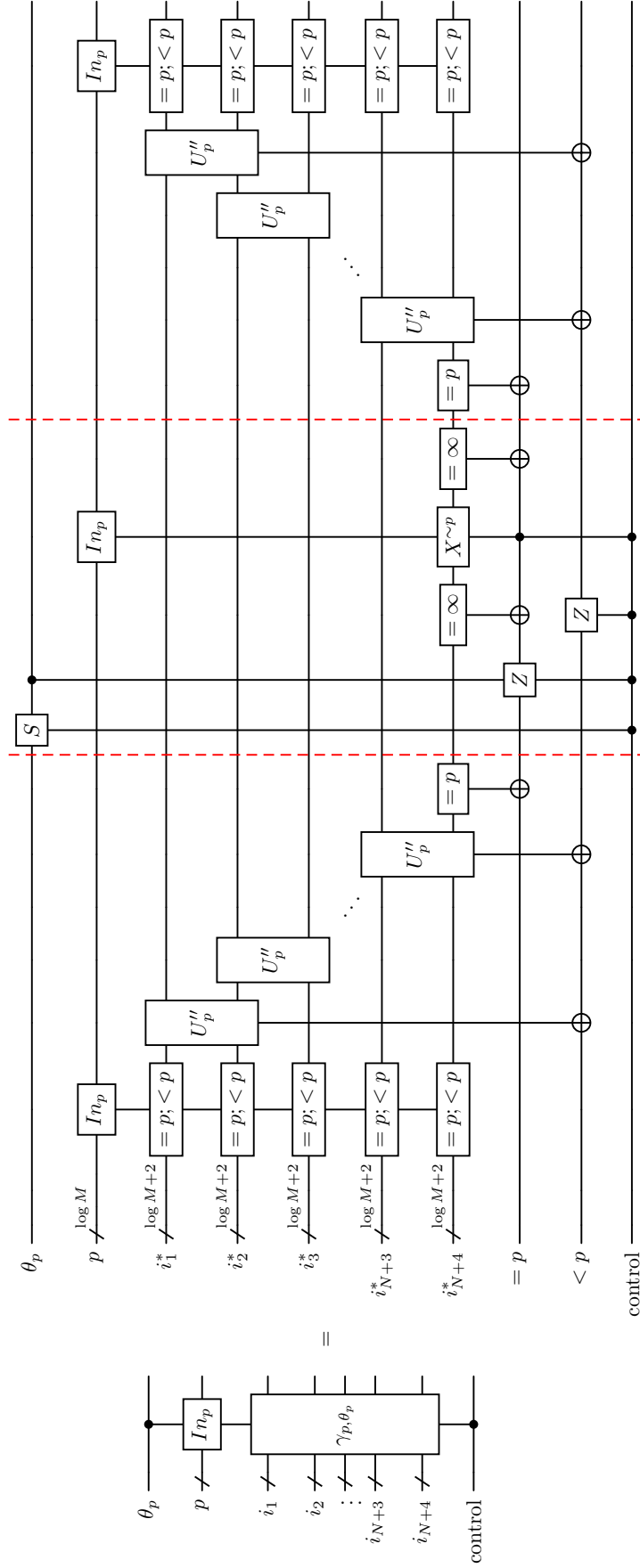


Figure A.19: The circuit implementing γ_{p,θ_p} . The bubble circuits U_p'' uses precalculated $= p$ and $< p$ information to reduce gate count and circuit depth. As a result, the registers i_n^* contain $\log M$ qubits storing the binary representation of the orbital indices, along with 2 additional ancilla qubits to store $= p$ and $< p$ data generated by the $= p; < p$ gates, totaling $\log M + 2$ qubits. The implementation of U_p'' is shown in Figure A.9. The dotted red line divides the circuit into three parts: BUBBLE, SWAP and BUBBLE † .

A.4.2. Cost Analysis

Similar to the trotterization circuits, we divide the γ_{p,θ_p} circuit into three parts as shown with the dotted red line of Figure A.19. The exact costings of such terms are shown in Table A.8 along with the costings of the γ_{p,θ_p} .

Table A.8: Costings of the gates of the γ_{p,θ_p} circuit of Figure A.19. Here, M is the number of orbitals and N is the total number of electrons.

Circuit	Costings
BUBBLE	$C^{\log M} X \quad N + 4$ $CSWAP \quad (N + 4) \log M - \log M + 2(N + 4) - 2$ $C^2 X \quad (N + 4) \log M + 2(N + 4) - 2$ $CX \quad 8(N + 4) \log M - \frac{1}{2}(N + 4) + \frac{1}{2}(N \bmod 2) - 5$ $X \quad \log M$
SWAP	$C^{\log M} X \quad 2$ $C^2 X \quad \log M + 3$ $CX \quad 1$ $H \quad 4$ $S \quad 1$
γ_{p,θ_p}	$C^{\log M} X \quad 2(N + 4) + 2$ $CSWAP \quad 2(N + 4) \log M + 4(N + 4) - 2 \log M - 4$ $C^2 X \quad 2(N + 4) \log M + 4(N + 4) + \log M - 1$ $CX \quad 16(N + 4) \log M - (N + 4) + (N \bmod 2) - 9$ $X \quad 2 \log M$ $H \quad 4$ $S \quad 1$

A.4.3. Plane-wave implementation

The PREPARE and SELECT circuits is adapted from the first-quantized plane-wave qubitization circuit [16] summarized in Section A.2 A.2.2 A.2.2.2. In the second-quantization, the Hamiltonian can be written as:

$$H^{(2)} = \hat{T}^{(2)} + \hat{U}^{(2)} + \hat{V}^{(2)}, \quad (\text{A.34})$$

$$\hat{T}^{(2)} = \sum_{p \in G} \frac{\|k_p\|^2}{2} a_p^\dagger a_p, \quad (\text{A.35})$$

$$\hat{U}^{(2)} = -\frac{4\pi}{\Omega} \sum_{\substack{\nu \in G \\ \nu \neq 0}} \sum_{\substack{p \in G \\ (p-\nu) \in G}} \left(\sum_{I=1}^{N_{at}} Z_I \frac{-e^{ik_\nu \cdot \mathbf{X}_I}}{\|k_\nu\|^2} \right) a_{p-\nu}^\dagger a_p, \quad (\text{A.36})$$

$$\hat{V}^{(2)} = \frac{2\pi}{\Omega} \sum_{\substack{i,j=1 \\ i \neq j}}^N \sum_{\substack{\nu \in G \\ \nu \neq 0}} \sum_{\substack{p,q \in G \\ (p+\nu) \in G \\ (q-\nu) \in G}} \frac{1}{\|k_\nu\|^2} a_{p+\nu}^\dagger a_{q-\nu}^\dagger a_q a_p. \quad (\text{A.37})$$

For the PREPARE circuit, ancilla d and e (see Table A.4) specifying the electron indices are removed. Instead, indices required for the Majorana operators (p, θ_p) , (q, θ_q) , (r, θ_r) , and (s, θ_s) are used. Here, we use the qubit label m to represent the qubit storing the orbital indices p, q, r, s and the label n to represent the qubit storing the parity indices $\theta_p, \theta_q, \theta_r, \theta_s$. The PREPARE circuit for the sorted-list encoding can be written in the following form, with each qubit

label explained in Table A.9.

$$\begin{aligned}
\text{PREPARE} &\rightarrow (\cos \theta |0\rangle + \sin \theta |1\rangle)_a |+\rangle_b \left(\frac{1}{\sqrt{3}} \sum_{w=0}^2 |w\rangle_f \right) \left(\frac{1}{2^{n_p-1}-1} \sum_{r,s=0}^{n_p-2} 2^{(r+s)/2} |r\rangle_g |s\rangle_h \right) \\
&\quad \left(\sqrt{\frac{\lambda_U}{\lambda_U + \lambda_V}} |0\rangle_i + \sqrt{\frac{\lambda_V}{\lambda_U + \lambda_V}} |1\rangle_i \right) \\
&\quad \left(\sqrt{\frac{p_\nu}{\lambda_\nu}} |0\rangle_j \sum_{\nu \in G_0} \frac{1}{\|\nu\|} |\nu\rangle_k + \sqrt{1-p_\nu} |1\rangle_j |\nu^\perp\rangle_k \right) \\
&\quad \left(\frac{1}{\sqrt{\sum_I Z_I}} \sum_{I=1}^{N_{at}} \sqrt{Z_I} |I\rangle_l \right) \text{INDICES} \tag{A.38} \\
\text{INDICES } |0\rangle_a &\rightarrow \left[\frac{1}{\sqrt{M}} \sum_{p \in G_0} |p\rangle |0\rangle |0\rangle |p\rangle \right]_m \left[\frac{1}{2} \sum_{\theta_p, \theta_q=0}^1 |\theta_p\rangle |0\rangle |0\rangle |\theta_q\rangle \right]_n \\
\text{INDICES } |1\rangle_a |0\rangle_i |\nu\rangle_k &\rightarrow \left[\frac{1}{\sqrt{M}} \sum_{p \in G_0} |p\rangle |0\rangle |0\rangle |p-\nu\rangle \right]_m \left[\frac{1}{2} \sum_{\theta_p, \theta_q=0}^1 |\theta_p\rangle |0\rangle |0\rangle |\theta_q\rangle \right]_n \\
\text{INDICES } |1\rangle_a |1\rangle_i |\nu\rangle_k &\rightarrow \left[\frac{1}{M} \sum_{p,q \in G_0} |p\rangle |q\rangle |q-\nu\rangle |p+\nu\rangle \right]_m \left[\frac{1}{4} \sum_{\theta_p, \theta_q, \theta_r, \theta_s=0}^1 |\theta_p\rangle |\theta_q\rangle |\theta_r\rangle |\theta_s\rangle \right]_n
\end{aligned}$$

Since it is assumed that M is a power of two [16], the INDICES operator can be implemented using Clifford gates. On the other hand, the SELECT circuit has the following form below, where γ_{p,θ_p} uses the circuit described in Section A.4 A.4.1.

$$\begin{aligned}
\text{SELECT}_{\hat{T}^{(2)}} &|b\rangle_b |w\rangle_f |r\rangle_g |s\rangle_h [|p\rangle |0\rangle |0\rangle |p\rangle]_m [|\theta_p\rangle |0\rangle |0\rangle |\theta_q\rangle]_n |\psi\rangle \\
&= (-1)^{b(p_w, r, p_w, s \oplus 1)} (-i)^{\theta_p} (i)^{\theta_q} |b\rangle_b |w\rangle_f |r\rangle_g |s\rangle_h [|p\rangle |0\rangle |0\rangle |p\rangle]_m [|\theta_p\rangle |0\rangle |0\rangle |\theta_q\rangle]_n \\
&\quad \times \gamma_{p,\theta_p} \gamma_{q,\theta_q} |\psi\rangle \\
\text{SELECT}_{\hat{U}^{(2)}} &|b\rangle_b |0\rangle_i |\nu\rangle_k |I\rangle_l [|p\rangle |0\rangle |0\rangle |p-\nu\rangle]_m [|\theta_p\rangle |0\rangle |0\rangle |\theta_q\rangle]_n |\psi\rangle \\
&= -e^{-ik_\nu \cdot \mathbf{X}_I} (-1)^{b(p-\nu) \notin G} (-i)^{\theta_p} (i)^{\theta_q} |b\rangle_b |0\rangle_i |\nu\rangle_k |I\rangle_l [|p\rangle |0\rangle |0\rangle |p-\nu\rangle]_m [|\theta_p\rangle |0\rangle |0\rangle |\theta_q\rangle]_n \\
&\quad \times \gamma_{p,\theta_p} \gamma_{q,\theta_q} |\psi\rangle \\
\text{SELECT}_{\hat{V}^{(2)}} &|b\rangle_b |1\rangle_i |\nu\rangle_k [|p\rangle |q\rangle |q-\nu\rangle |p+\nu\rangle]_m [|\theta_p\rangle |\theta_q\rangle |\theta_r\rangle |\theta_s\rangle]_n |\psi\rangle \\
&= (-1)^{b([(p+\nu) \notin G] \vee [(q-\nu) \notin G])} (-i)^{\theta_p+\theta_q} (i)^{\theta_r+\theta_s} |b\rangle_b |1\rangle_i |\nu\rangle_k [|p\rangle |q\rangle |q-\nu\rangle |p+\nu\rangle]_m \\
&\quad \times [|\theta_p\rangle |\theta_q\rangle |\theta_r\rangle |\theta_s\rangle]_n \gamma_{p,\theta_p} \gamma_{q,\theta_q} \gamma_{r,\theta_r} \gamma_{s,\theta_s} |\psi\rangle
\end{aligned}$$

Table A.9: Legend for the qubit label of Equation A.38.

Qubit Label	Description
a	$ 0\rangle_a$ tells the SELECT circuit to apply $\hat{T}^{(2)}$ $ 1\rangle_a$ tells the SELECT circuit to apply $\hat{U}^{(2)} + \hat{V}^{(2)}$
b	Used to cancel out the unitary application when $(p + \nu) \notin G$ and when $(q - \nu) \notin G$ for $\hat{U}^{(2)}$ and $\hat{V}^{(2)}$ (see Equation A.36 and A.37)
c	Specifies if $i = j$ for the case of $\hat{V}^{(2)}$ (see Equation A.37)
w	$ 0\rangle_w, 1\rangle_w, 2\rangle_w$ specifies the Cartesian axes x, y , and z , respectively, used for the SELECT circuit to apply $\hat{T}^{(2)}$
g and h	Used for the application of $\ k_p\ ^2$ in $\hat{T}^{(2)}$ (see Equation A.35)
i	$ 1\rangle_a 0\rangle_i$ tells the SELECT circuit to apply $\hat{U}^{(2)}$ $ 1\rangle_a 1\rangle_i$ tells the SELECT circuit to apply $\hat{V}^{(2)}$
j and k	Creates the $1/\ k_\nu\ ^2$ coefficients for $\hat{U}^{(2)}$ and $\hat{V}^{(2)}$ (see Equation A.36 and A.37)
l	The ion indices I for $\hat{U}^{(2)}$ (see Equation A.36)
m	The orbital indices p, q, r, s for the Majorana operators
n	The parity indices $\theta_p, \theta_q, \theta_r, \theta_s$ for the Majorana operators

A.4.4. Cost Estimates on Model Systems

Table A.10: Costings of the select operator along with the qubitization operator for molecular systems with MO basis sets using $\text{PREPARE}_{\text{SF}}$. Here, QROM with dirty ancilla assistance is used to minimize the number of ancilla qubits. The parameters of the FeMoCo calculations are obtained from [6]. Toffoli costs are converted to T gates with 2 T gates for every Toffoli gate [28].

System	Basis set	Orbs	QPE steps	SELECT		PREPARE		TOTAL	
				Qubits	T gates/Step	Qubits	T gates/Step	Qubits	T gates/Step
H ₂ O 10 elec Model System	STO-3G	7	17	87	3.3×10^3	123	1.7×10^3	210	5.0×10^3
	6-311G	19	20	115	4.7×10^3	153	4.5×10^3	268	9.2×10^3
	cc-pvdz	24	20	115	4.7×10^3	161	5.6×10^3	276	1.0×10^4
	cc-pvtz	58	24	129	5.4×10^3	186	1.4×10^4	315	1.9×10^4
	cc-pvqz	115	27	143	6.1×10^3	210	3.1×10^4	353	3.7×10^4
	cc-pv5z	201	30	157	6.7×10^3	234	7.1×10^4	391	7.7×10^4
CO ₂ 22 elec Model System	STO-3G	15	19	185	7.5×10^3	141	3.3×10^3	326	1.1×10^4
	6-311G	39	22	237	1.0×10^4	171	8.8×10^3	408	1.9×10^4
	cc-pvdz	42	22	237	1.0×10^4	174	9.7×10^3	411	2.0×10^4
	cc-pvtz	90	25	263	1.1×10^4	203	2.3×10^4	466	3.4×10^4
	cc-pvqz	165	28	289	1.2×10^4	222	5.2×10^4	511	6.4×10^4
	cc-pv5z	273	30	315	1.4×10^4	241	1.1×10^5	556	1.3×10^5
EC 46 elec Battery Materials	STO-3G	34	21	453	1.9×10^4	169	7.4×10^3	622	2.7×10^4
	6-311G	90	26	503	2.2×10^4	204	2.1×10^4	707	4.3×10^4
	cc-pvdz	104	27	503	2.2×10^4	209	2.5×10^4	712	4.6×10^4
	cc-pvtz	236	31	553	2.4×10^4	238	6.5×10^4	791	8.9×10^4
LiPF ₆ 72 elec Battery Materials	STO-3G	44	23	687	2.9×10^4	176	9.1×10^3	863	3.8×10^4
	6-311G	112	27	763	3.3×10^4	209	2.5×10^4	972	5.8×10^4
	cc-pvdz	116	27	763	3.3×10^4	209	2.7×10^4	972	6.0×10^4
	cc-pvtz	244	31	839	3.6×10^4	237	5.8×10^4	1076	9.4×10^4
FeMoCo (from [6])	RWSWT (54 elec)	54	23	525	2.2×10^4	178	1.0×10^4	703	3.3×10^4
	LLDUC (113 elec)	76	22	1173	5.1×10^4	176	1.3×10^4	1349	6.4×10^4

Table A.11: Costings of the select operator along with the qubitization operator for molecular systems with MO basis sets using $\text{PREPARE}_{\text{SF}}$. Here, QROM with clean ancilla assistance is used to minimize the number of Toffolis. The parameters of the FeMoCo calculations are obtained from [6]. The clean ancillae of the PREPARE step can be reused for the SELECT step. As a result, the qubit cost of the SELECT steps only contains data qubits $N \log M$ as opposed to $(N + 4)(\log M + 2) + 3$. Toffoli costs are converted to T gates with 2 T gates for every Toffoli gate [28].

System	Basis Set	Orbs	QPE steps	SELECT		PREPARE		TOTAL	
				Qubits	Toffoli/Step	Qubits	Toffoli/Step	Qubits	Toffoli/Step
H ₂ O 10 elec Model System	STO-3G	7	17	40	3.3×10^3	160	1.3×10^3	200	4.7×10^3
	6-311G	19	20	60	4.7×10^3	213	2.8×10^3	273	7.5×10^3
	cc-pvdz	24	20	60	4.7×10^3	284	3.3×10^3	344	8.0×10^3
	cc-pvtz	58	24	70	5.4×10^3	473	6.7×10^3	543	1.2×10^4
	cc-pvqz	115	27	80	6.1×10^3	864	1.2×10^4	944	1.8×10^4
	cc-pv5z	201	30	90	6.7×10^3	1703	2.1×10^4	1793	2.8×10^4
CO ₂ 22 elec Model System	STO-3G	15	19	110	7.5×10^3	204	2.2×10^3	314	9.6×10^3
	6-311G	39	22	154	1.0×10^4	307	4.7×10^3	461	1.5×10^4
	cc-pvdz	42	22	154	1.0×10^4	310	5.2×10^3	464	1.5×10^4
	cc-pvtz	90	25	176	1.1×10^4	522	9.9×10^3	698	2.1×10^4
	cc-pvqz	165	28	198	1.2×10^4	924	1.7×10^4	1122	3.0×10^4
	cc-pv5z	273	30	220	1.4×10^4	1774	2.8×10^4	1994	4.2×10^4
EC 46 elec Battery Materials	STO-3G	34	21	322	1.9×10^4	305	4.1×10^3	627	2.3×10^4
	6-311G	90	26	368	2.2×10^4	524	1.0×10^4	892	3.2×10^4
	cc-pvdz	104	27	368	2.2×10^4	864	1.1×10^4	1232	3.3×10^4
	cc-pvtz	236	31	414	2.4×10^4	1740	2.5×10^4	2154	4.9×10^4
LiPF ₆ 72 elec Battery Materials	STO-3G	44	23	504	2.9×10^4	343	4.9×10^3	847	3.4×10^4
	6-311G	112	27	576	3.3×10^4	864	1.2×10^4	1440	4.5×10^4
	cc-pvdz	116	27	576	3.3×10^4	864	1.2×10^4	1440	4.5×10^4
	cc-pvtz	244	31	648	3.6×10^4	1740	2.5×10^4	2388	6.2×10^4
FeMoCo (from [6])	RWSWT (54 elec)	54	23	378	2.2×10^4	457	5.4×10^3	835	2.8×10^4
	LLDUC (113 elec)	76	22	904	5.1×10^4	464	6.5×10^3	1368	5.7×10^4

A.4.5. Majorana-free Implementations

The mechanisms of the Majorana-free select circuits are similar to the Majorana-free trotterization circuit. Given an arbitrary Slater determinant $|\psi\rangle$, the Hermitian operator $a_p^\dagger a_q + a_q^\dagger a_p$ should swap the occupation of p and q if either p is occupied or q is occupied, but not both, while applying the appropriate phase depending on the parity of the wavefunction. Otherwise, $|\psi\rangle$ should vanish. The former requirement is implemented using similar techniques as the Majorana-free Trotterization circuit, while the latter requirement uses an ancilla qubit in the $|+\rangle$ state as done in [16]. If the requirements are not satisfied, a Z gate is applied in this $|+\rangle$ state, followed by uncomputation. This is done similarly for the other Hermitian operators $ia_p^\dagger a_q - ia_q^\dagger a_p$, $a_p^\dagger a_q^\dagger a_r a_s + a_s^\dagger a_r^\dagger a_q a_p$, and $ia_p^\dagger a_q^\dagger a_r a_s - ia_s^\dagger a_r^\dagger a_q a_p$. As a result, the PREPARE circuit will simply need to prepare an additional ancilla qubit in the $|+\rangle$ state.

$$\begin{aligned} |+\rangle_b |p\rangle |q\rangle |\psi\rangle &\rightarrow (-1)^{b[(p \in \psi) \oplus (q \in \psi) \oplus 1]} |+\rangle_a |p\rangle |q\rangle |\psi\rangle \\ &\rightarrow \begin{cases} |+\rangle_b |p\rangle |q\rangle a_p^\dagger a_q |\psi\rangle & \text{if } (p \in \psi) \oplus (q \in \psi) \\ |-\rangle_b |p\rangle |q\rangle a_p^\dagger a_q |\psi\rangle & \text{otherwise} \end{cases} \end{aligned}$$

The circuit to implement $a_p^\dagger a_q + a_q^\dagger a_p$ is shown in Figure A.20 while the circuit to implement $a_p^\dagger a_q^\dagger a_r a_s + a_s^\dagger a_r^\dagger a_q a_p$ is shown in Figure A.21. Those circuits can be trivially modified to obtain implementations for $ia_p^\dagger a_q - ia_q^\dagger a_p$ and $ia_p^\dagger a_q^\dagger a_r a_s - ia_s^\dagger a_r^\dagger a_q a_p$. Moreover, the use of the $|+\rangle$ operator can be extended to implement the non-Hermitian variants $a_p^\dagger a_q$ and $a_p^\dagger a_q^\dagger a_r a_s$.

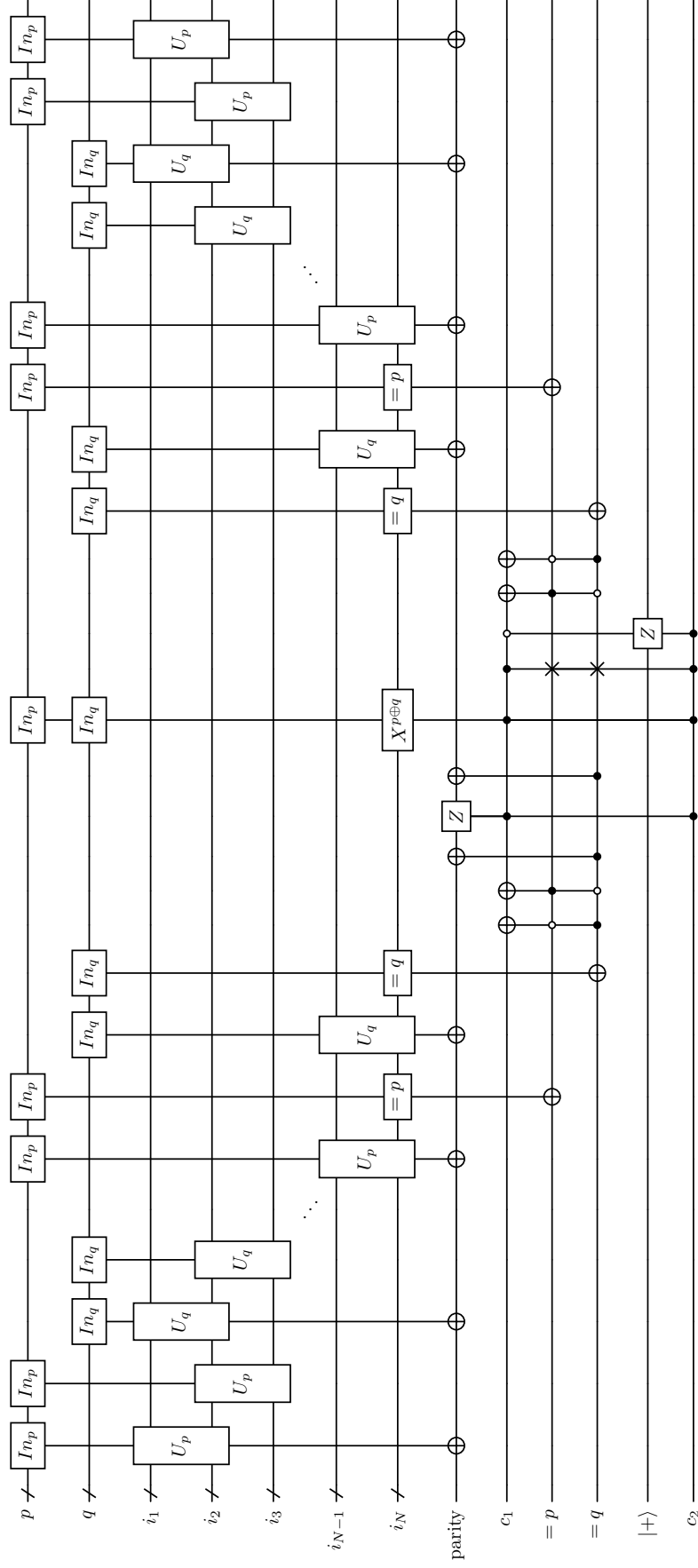


Figure A.20: Circuit to implement $a_p^\dagger a_q + a_q^\dagger a_p$. The circuit can be trivially modified to obtain $ia_p^\dagger a_q - ia_q^\dagger a_p$.

Figure A.21: Circuit to implement $a_p^\dagger a_q^\dagger a_r a_s + a_s^\dagger a_r^\dagger a_q a_p$. The circuit can be trivially modified to obtain $ia_p^\dagger a_q^\dagger a_r a_s - ia_s^\dagger a_r^\dagger a_q a_p$.

Appendix A.5: Measurement circuits for the Sorted-List Encoding

Here, we show how to measure the expectation value of the Hermitian fermionic operators in the sorted-list encoding. Before that, we introduce the following notation for the Slater determinant. Let p, q, r, s be the orbital indices, we then write $|\Psi_p^q\rangle$ to denote an arbitrary Slater determinant where orbital p is unoccupied while orbital q is occupied. Similarly, we write $|\Psi_{pq}^{rs}\rangle$ to denote an arbitrary Slater determinant where both orbitals p and q are unoccupied and orbitals r and s are occupied.

First, we show the procedure for measuring

$$\langle a_p^\dagger a_p \rangle \quad \text{and} \quad \langle a_p^\dagger a_q^\dagger a_q a_p \rangle.$$

Similar to the measurement of Z gates for the case of Pauli strings, one would simply measure the wavefunction. The expectation value of the above terms would then be equal to the probability of having p in one of the registers for the case of $\langle a_p^\dagger a_p \rangle$ and having p and q in two of the registers for the case of $\langle a_p^\dagger a_q^\dagger a_q a_p \rangle$. Optionally, $= p$ and $= q$ gates can be added to every register, targeting one ancilla qubit for the $= p$ gates and another for the $= q$ gates. This way, only the ancilla qubits would be measured.

Next, we move on to the measurement of $\langle a_p^\dagger a_q + a_q^\dagger a_p \rangle$. Similarly to the measurement of the X gate, the eigenvalues of these operators are $(\Psi_p^q \pm \Psi_q^p)/\sqrt{2}$. However, the eigenvalue of $(\Psi_p^q + \Psi_q^p)/\sqrt{2}$ is not necessarily $+1$. The sign of the eigenvalue can be flipped depending on the parity of the fermionic operators. Assuming that $p > q$, the parity of $a_p^\dagger a_q$ and $a_q^\dagger a_p$ can be extracted from Equation A.26.

$$\begin{aligned} \text{PARITY}\{a_p^\dagger a_q\} &= \Theta_{pq} \left(\prod_{i=q}^{p-1} Z_i \right) = - \prod_{i=q}^{p-1} Z_i = \prod_{i=q+1}^{p-1} Z_i, \\ \text{PARITY}\{a_q^\dagger a_p\} &= \Theta_{qp} \left(\prod_{i=q}^{p-1} Z_i \right) = \prod_{i=q}^{p-1} Z_i = \prod_{i=q+1}^{p-1} Z_i, \\ \text{PARITY}\{a_p^\dagger a_q\} &= \text{PARITY}\{a_q^\dagger a_p\}. \end{aligned}$$

The last equality of the $\text{PARITY}\{a_p^\dagger a_q\}$ equation is because q must be occupied otherwise $a_p^\dagger a_q |\psi\rangle$ will vanish. The same reason can be applied for the last equality of $\text{PARITY}\{a_q^\dagger a_p\}$. As a result, we have the following eigenvalue-eigenvector pair:

$$(a_p^\dagger a_q + a_q^\dagger a_p)(\Psi_p^q \pm \Psi_q^p) = \pm \text{PARITY}\{a_p^\dagger a_q\}(\Psi_p^q \pm \Psi_q^p).$$

The measurement circuit for this is shown in Figure A.22. It would first perform parity measurements similar to the Trotterization circuits, before transforming $(\Psi_p^q + \Psi_q^p)$ to $|0\rangle_p$ and $(\Psi_p^q - \Psi_q^p)$ to $|1\rangle_p$ in the ancilla labeled “ p ”. Measurements would then involve obtaining the weighted probability distribution of states as follows

$$\begin{aligned} \langle a_p^\dagger a_q + a_q^\dagger a_p \rangle &= \text{Prob}(|0\rangle_{\text{parity}} |0\rangle_p |1\rangle_q) - \text{Prob}(|0\rangle_{\text{parity}} |1\rangle_p |1\rangle_q) \\ &\quad - \text{Prob}(|1\rangle_{\text{parity}} |0\rangle_p |1\rangle_q) + \text{Prob}(|1\rangle_{\text{parity}} |1\rangle_p |1\rangle_q). \end{aligned} \quad (\text{A.39})$$

In this case, $\text{Prob}(|0\rangle_{\text{parity}} |0\rangle_p |1\rangle_q)$ is the probability of obtaining the state where the parity qubit is 0, qubit p is 0 and qubit q is 1. The rightmost CNOT and H gates of Figure A.22 acts on qubit p and q as follows:

$$\begin{aligned} \frac{1}{\sqrt{2}}(|\Psi_p^q\rangle \pm |\Psi_q^p\rangle) &\Rightarrow \frac{1}{\sqrt{2}}[|0\rangle_p |1\rangle_q \pm |1\rangle_p |0\rangle_q] \\ &\xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}}[|0\rangle_p |1\rangle_q \pm |1\rangle_p |1\rangle_q] = |\pm\rangle_p |1\rangle_q \\ &\xrightarrow{H} |0/1\rangle_p |1\rangle_q. \end{aligned}$$

For the cases where neither p nor q is occupied ($|0\rangle_p |0\rangle_q$) or when both p and q are occupied ($|1\rangle_p |1\rangle_q$), the rightmost CNOT gate would result in $|0\rangle_q$, which are excluded from the expectation value calculation of Equation A.39.

We then move on to the measurement of $\langle ia_p^\dagger a_q - ia_q^\dagger a_p \rangle$. In this case, we have the following eigenvalue-eigenvector pair:

$$(ia_p^\dagger a_q - ia_q^\dagger a_p)(\Psi_p^q \pm i\Psi_q^p) = \pm \text{PARITY}\{a_p^\dagger a_q\}(\Psi_p^q \pm i\Psi_q^p).$$

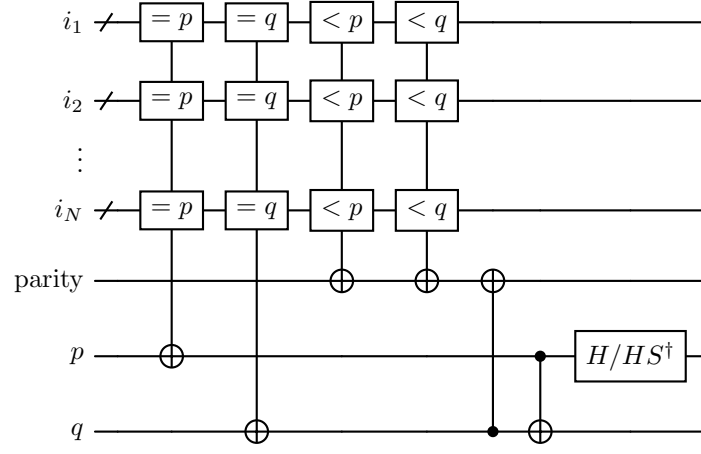


Figure A.22: Circuit to measure $\langle a_p^\dagger a_q + a_q^\dagger a_p \rangle$ where the H gate is used, and $\langle ia_p^\dagger a_q - ia_q^\dagger a_p \rangle$ where the $HS^\dagger$ gate is used.

Consequently, we would use the $HS^\dagger$ gate instead of the H gate for Figure A.22. The circuit would then act as follows:

$$\begin{aligned}
 \frac{1}{\sqrt{2}}(|\Psi_p^q\rangle \pm i|\Psi_q^p\rangle) &\Rightarrow \frac{1}{\sqrt{2}}[|0\rangle_p |1\rangle_q \pm i|1\rangle_p |0\rangle_q] \\
 &\xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}}[|0\rangle_p |1\rangle_q \pm i|1\rangle_p |1\rangle_q] \\
 &\xrightarrow{S^\dagger} \frac{1}{\sqrt{2}}[|0\rangle_p |1\rangle_q \pm |1\rangle_p |1\rangle_q] = |\pm\rangle_p |1\rangle_q \\
 &\xrightarrow{H} |0/1\rangle_p |1\rangle_q.
 \end{aligned}$$

After measurement, Equation A.39 can again be used for $\langle ia_p^\dagger a_q - ia_q^\dagger a_p \rangle$.

The measurements of $\langle a_p^\dagger a_q^\dagger a_q a_s + a_s^\dagger a_q^\dagger a_q a_p \rangle$ and $\langle ia_p^\dagger a_q^\dagger a_q a_s - ia_s^\dagger a_q^\dagger a_q a_p \rangle$ are performed in a similar way to the 1-RDM terms, except that the $= q$ check is introduced either in the circuit or during the measurement process. Finally, the measurements of $\langle a_p^\dagger a_q^\dagger a_r a_s + a_s^\dagger a_r^\dagger a_q a_p \rangle$ and $\langle ia_p^\dagger a_q^\dagger a_r a_s - ia_s^\dagger a_r^\dagger a_q a_p \rangle$ are performed using the circuit of Figure A.23, with a measurement procedure identical to the 1-RDM terms, except that the check for the occupation of p is replaced with checking the occupation of pq and checking the occupation of q is replaced with checking the occupation of rs .

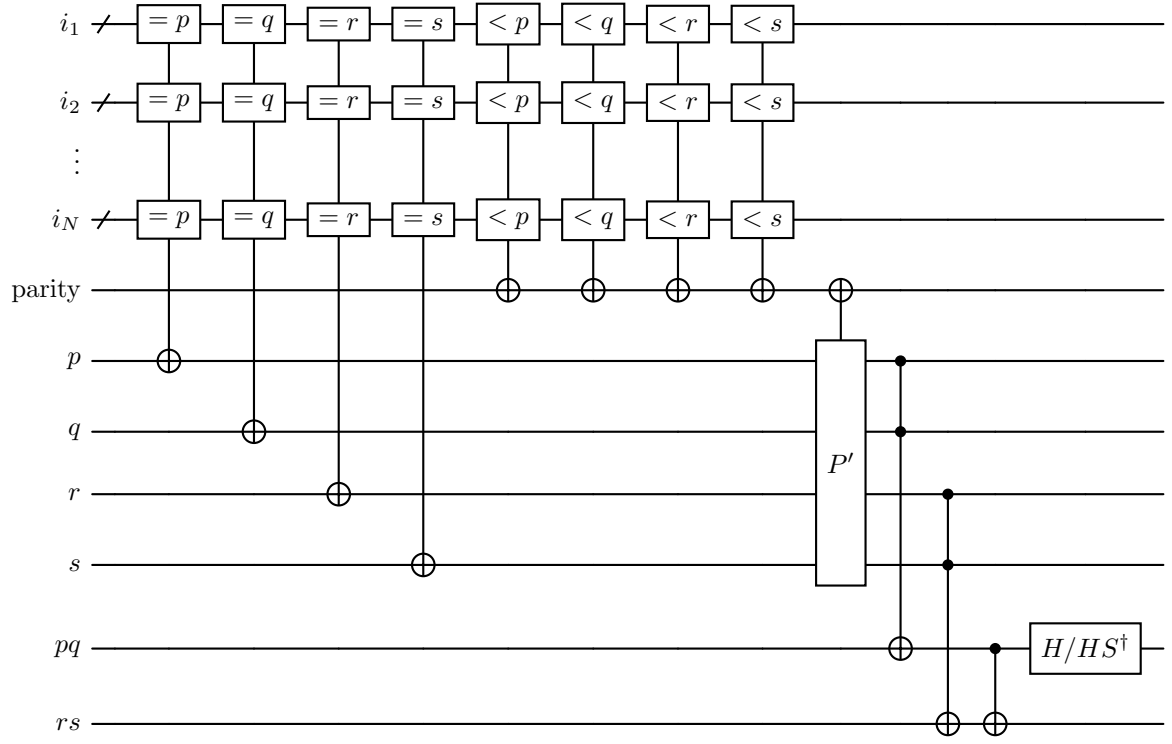


Figure A.23: Circuit to measure $\langle a_p^\dagger a_q^\dagger a_r a_s + a_s^\dagger a_r^\dagger a_q a_p \rangle$ where the H gate is used, and $\langle i a_p^\dagger a_q^\dagger a_r a_s - i a_s^\dagger a_r^\dagger a_q a_p \rangle$ where the $HS^\dagger$ gate is used.